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Year II — Mathematical Methods

Based on lectures by Various Lecturers

Notes taken by Anonymous

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Part I

Vector Calculus

Chapter 1

Derivatives and Coordinates

1.1 Vector and Scalar Functions

1.1.1 Vector Functions

Definition 1.1.1 (Vector Function)**.** A *vector function* f is a function of the form:

$$f : \mathbb{R} \to \mathbb{R}^n$$

which takes a numerical input and returns a vector

Definition 1.1.2 (Derivative of a Vector Function)**.** A vector function $\mathbf{F}(x)$ is differentiable if

$$\delta\mathbf{F} \stackrel{\text{def}}{=} \mathbf{F}(x+\delta x) - \mathbf{F}(x) = \mathbf{F}'(x)\delta x + o(\delta x)$$

for some $\mathbf{F}'(x)$.$\mathbf{F}'(x)$ is called the derivative of $\mathbf{F}(x)$. This is the equivalent of writing:

$$\mathbf{F}' = \frac{\mathrm{d}\mathbf{F}}{\mathrm{d}x} = \lim_{\delta x \to 0} \frac{1}{\delta x}[\mathbf{F}(x+\delta x) - \mathbf{F}(x)]$$

Example 1.1.3.

$$\mathbf{r}(t) = \begin{pmatrix} t^2 \\ e^t \\ \sin t \end{pmatrix}$$

Differentiate componentwise:

$$\mathbf{r}'(t) = \begin{pmatrix} 2t \\ e^t \\ \cos t \end{pmatrix}$$

At $t = 0$:

$$\mathbf{r}'(0) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Proposition 1.1.4 (Componentwise differentiation)**.** Using differential notation, the differentiability condition can be written as

$$\mathrm{d}\mathbf{F} = \mathbf{F}'(x)\mathrm{d}x$$

Given a basis $\mathbf{e}_i$ that is independent of x, vector differentiation is performed component wise:

$$\mathbf{F}'(x) = F_i'(x)\mathbf{e}_i.$$

Proposition 1.1.5 (Vector Differentiation on Operations)**.**

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{fg}) &= \frac{\mathrm{d}\mathbf{f}}{\mathrm{d}t}\mathbf{g} + \mathbf{f}\frac{\mathrm{d}\mathbf{g}}{\mathrm{d}t} \\ \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{g}\cdot\mathbf{h}) &= \frac{\mathrm{d}\mathbf{g}}{\mathrm{d}t}\cdot\mathbf{h} + \mathbf{g}\cdot\frac{\mathrm{d}\mathbf{h}}{\mathrm{d}t} \\ \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{g}\times\mathbf{h}) &= \frac{\mathrm{d}\mathbf{g}}{\mathrm{d}t}\times\mathbf{h} + \mathbf{g}\times\frac{\mathrm{d}\mathbf{h}}{\mathrm{d}t} \end{aligned}$$

We note here that the order of operations matters for the case of the cross product since it is anti symmetric

1.1.2 Scalar Fields

Definition 1.1.6 (Scalar Field)**.** A *scalar field* f is a mapping of the form:

$$f : \mathbb{R}^n \to \mathbb{R}$$

which takes a vector and returns a numerical output.

Remark 1.1.7 (Distinction between Vector and Scalar Function)**.** A scalar function takes a position and returns a numerical output (example : potential at a point) where a vector function takes a number and maps it to a vector (example : the velocity associated to a particle at t)

Definition 1.1.8 (Limit of a Vector)**.** The limit of vectors is defined using the norm. So $\mathbf{v} \to \mathbf{c}$ iff $|\mathbf{v} - \mathbf{c}| \to 0$. So a vector approaches another vector if and only if the difference between them goes to 0. Similarly, $f(\mathbf{r}) = o(\mathbf{r})$ means $\frac{|f(\mathbf{r})|}{|\mathbf{r}|} \to 0$ as $\mathbf{r} \to \mathbf{0}$.

Definition 1.1.9 (Gradient / Derivative of a Scalar Function)**.** A scalar function $f(\mathbf{r})$ is differentiable at $\mathbf{r}$ if

$$\delta f \overset{\text{def}}{=} f(\mathbf{r}+\delta\mathbf{r}) - f(\mathbf{r}) = (\nabla f)\cdot\delta\mathbf{r} + o(\delta\mathbf{r})$$

for some vector ∇f, the gradient of f at $\mathbf{r}$.

Example 1.1.10.

$$f(x,y) = x^2 y + \sin y$$

The derivative is the gradient row/column (depending on convention). The partial derivatives are: - $\frac{\partial f}{\partial x} = 2xy$ - $\frac{\partial f}{\partial y} = x^2 + \cos y$

Gradient:

$$\nabla f(x,y) = \begin{pmatrix} 2xy \\ x^2 + \cos y \end{pmatrix}$$

At $(1,0)$:

$$\nabla f(1,0) = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

Remark 1.1.11 (Importance of Taylor Derivative Definition)**.** We see here the importance of this new way of defining a derivative in higher dimensions since dividing by a vector makes no sense if we were to try and rearrange.

Definition 1.1.12 (Directional Derivative)**.** The directional derivative of f along $\mathbf{n}$ is

$$\mathbf{n}\cdot\nabla f = \lim_{h\to 0}\frac{1}{h}[f(\mathbf{r}+h\mathbf{n}) - f(\mathbf{r})],$$

It refers to how fast f changes when we move in the direction of $\mathbf{n}$.

Remark 1.1.13 (Maximising the Directional Derivative)**.** When the derivative moves in the direction of $\mathbf{n}$ then we have the maximum possible value of the directional derivative and thus points in the direction of the greatest slope

Theorem 1.1.14 (Gradient expressed in Partial Derivative)**.** The gradient is

$$\nabla f = \frac{\partial f}{\partial x_i}\mathbf{e}_i$$

Hence we can write the condition of differentiability as

$$\delta f = \frac{\partial f}{\partial x_i}\delta x_i + o(\delta \mathbf{x}).$$

In differential notation, we write

$$\mathrm{d}f = \nabla f \cdot \mathrm{d}\mathbf{r} = \frac{\partial f}{\partial x_i}\,\mathrm{d}x_i,$$

which is the chain rule for partial derivatives.

Theorem 1.1.15 (Chain Rule for a Scalar Function)**.** Given a function $f(\mathbf{r}(u))$,

$$\frac{\mathrm{d}f}{\mathrm{d}u} = \nabla f \cdot \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}u} = \frac{\partial f}{\partial x_i}\frac{\mathrm{d}x_i}{\mathrm{d}u}.$$

Note that if we drop the $\mathrm{d}u$, we simply get

$$\mathrm{d}f = \nabla f \cdot \mathrm{d}\mathbf{r} = \frac{\partial f}{\partial x_i}\,\mathrm{d}x_i,$$

which is what we've previously had.

1.1.3 Vector Fields

Definition 1.1.16 (Vector Field)**.** A *vector field* f is a function of the form:

$$f : \mathbb{R}^n \to \mathbb{R}^m$$

which takes a vector input of dimension n and returns a vector of dimension m

Definition 1.1.17 (Derivative of a Vector Field)**.** A vector field $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable if

$$\delta\mathbf{F} \overset{\text{def}}{=} \mathbf{F}(\mathbf{x}+\delta\mathbf{x}) - \mathbf{F}(\mathbf{x}) = M\delta\mathbf{x} + o(\delta\mathbf{x})$$

for some $m \times n$ matrix M.M is the derivative of $\mathbf{F}$.

Theorem 1.1.18 (Derivative of a Vector field in ij notation)**.** Given an arbitrary function $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$ that maps $\mathbf{x} \mapsto \mathbf{y}$ and a choice of basis, we can write $\mathbf{F}$ as a set of m functions $y_j = F_j(\mathbf{x})$ such that $\mathbf{y} = (y_1, y_2, \cdots, y_m)$. Then

$$\mathrm{d}y_j = \frac{\partial F_j}{\partial x_i}\,\mathrm{d}x_i.$$

and we can write the derivative as

$$M_{ji} = \frac{\partial y_j}{\partial x_i}.$$

Note that we could have used this as the definition of the derivative. However, the original definition is superior because it does not require a selection of coordinate system.

Example 1.1.19. Take the vector field

$$F(x,y) = \begin{pmatrix} x^2 y \\ \sin(xy) \end{pmatrix}.$$

The derivative matrix is obtained by differentiating each component with respect to each variable.

$$DF(x,y) = \begin{pmatrix} \frac{\partial}{\partial x}(x^2y) & \frac{\partial}{\partial y}(x^2y) \\ \frac{\partial}{\partial x}(\sin(xy)) & \frac{\partial}{\partial y}(\sin(xy)) \end{pmatrix}.$$

Computing the entries: - $\frac{\partial}{\partial x}(x^2y) = 2xy$ - $\frac{\partial}{\partial y}(x^2y) = x^2$ - $\frac{\partial}{\partial x}(\sin(xy)) = y\cos(xy)$ - $\frac{\partial}{\partial y}(\sin(xy)) = x\cos(xy)$

So

$$DF(x,y) = \begin{pmatrix} 2xy & x^2 \\ y\cos(xy) & x\cos(xy) \end{pmatrix}$$

Example at the point $(1,2)$:

$$DF(1,2) = \begin{pmatrix} 4 & 1 \\ 2\cos(2) & \cos(2) \end{pmatrix}.$$

Definition 1.1.20 (Smooth Function)**.** A function is smooth if it can be differentiated any number of times. This requires that all partial derivatives exist and are totally symmetric in i, j and k (i.e. the differential operator is commutative).

Theorem 1.1.21 (Chain Rule of Vector Fields)**.** Suppose $g : \mathbb{R}^p \to \mathbb{R}^n$ and $f : \mathbb{R}^n \to \mathbb{R}^m$. Suppose that the coordinates of the vectors in $\mathbb{R}^p, \mathbb{R}^n$ and $\mathbb{R}^m$ are u_a, x_i and y_r respectively. By the chain rule,

$$\frac{\partial y_r}{\partial u_a} = \frac{\partial y_r}{\partial x_i}\frac{\partial x_i}{\partial u_a},$$

with summation implied. Writing in matrix form,

$$M(f \circ g)_{ra} = M(f)_{ri} M(g)_{ia}.$$

Alternatively, in operator form,

$$\frac{\partial}{\partial u_a} = \frac{\partial x_i}{\partial u_a}\frac{\partial}{\partial x_i}.$$

1.2 Inverse Functions

Theorem 1.2.1 (Derivative of an Inverse Function in Vector Field)**.** Suppose $g, f : \mathbb{R}^n \to \mathbb{R}^n$ are inverse functions, i.e. $g \circ f = f \circ g = \mathrm{id(x)}$. This is where $\mathrm{id}(\mathbf{x}) = \mathbf{x}$ Suppose that $f(\mathbf{x}) = \mathbf{u}$ and $g(\mathbf{u}) = \mathbf{x}$.

Since the derivative of the identity function $\mathrm{id}(\mathbf{x})$ is the identity matrix (if you differentiate $\mathbf{x}$ wrt to $\mathbf{x}$, you get 1), we must have

$$M(f \circ g) = I.$$

Therefore we know that

$$M(g) = M(f)^{-1}.$$

We derive this result more formally by noting

$$\frac{\partial u_b}{\partial u_a} = \delta_{ab}.$$

So by the chain rule,

$$\frac{\partial u_b}{\partial x_i}\frac{\partial x_i}{\partial u_a} = \delta_{ab},$$

i.e. $M(f \circ g) = I$.

Remark 1.2.2 (Determinants)**.** For the inverse functions f and g:

$$\det M(f) \det M(g) = 1$$

1.3 Coordinate Systems

Cylindrical polars	**Spherical polars**
Conversion formulae	
$x_1 = \rho\cos\varphi$ $x_2 = \rho\sin\varphi$ $x_3 = z$	$x_1 = r\sin\theta\cos\varphi$ $x_2 = r\sin\theta\sin\varphi$ $x_3 = r\cos\theta$
Basis vectors	
$\mathbf{e}_\rho = (\cos\varphi, \sin\varphi, 0)$ $\mathbf{e}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ $\mathbf{e}_z = (0, 0, 1)$	$\mathbf{e}_r = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$ $\mathbf{e}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ $\mathbf{e}_\theta = (\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta)$

Chapter 2

Line Integrals

2.1 Parametrisation and Curves

Definition 2.1.1 (Curve). A curve C is a 1-d subset of $\mathbb{R}^n$, for $n > 1$.

It will be more use to us to define a curve through parametrisation.

Definition 2.1.2 (Parametrisation). *Parametrisation* is a function, $\boldsymbol{x}(t)$, which maps a real interval $[t_0, t_1]$ into $\mathbb{R}^n$. This is a vector-valued function

$$\boldsymbol{x}(t) = x_1(t)\boldsymbol{e}_1 + x_2(t)\boldsymbol{e}_2 + \ldots + x_n(t)\boldsymbol{e}_n,$$

where $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n\}$ denote the Cartesian unit vectors. When $\boldsymbol{n = 2}$ or $\boldsymbol{n = 3}$, we will usually write $x_1 \equiv x, x_2 \equiv y$ and $x_3 \equiv z$.

Definition 2.1.3 (Closed and Simple Curve). If $\boldsymbol{x}(t_1) = \boldsymbol{x}(t_0)$ then the curve is called closed. A curve that does not cross itself is called simple.

Remark 2.1.4. A common parametrisation for a circle is polar coordinates. We note here however that parametrisation is not unique and a single curve may be described by multiple parametrisation.

Definition 2.1.5 (Differentaible Parametrisation). A parametrisation $\boldsymbol{x}(t)$ is differentiable if it has a well-defined tangent vector,

$$\frac{d\boldsymbol{x}}{dt} = \lim_{dt \to 0} \frac{\boldsymbol{x}(t + dt) - \boldsymbol{x}(t)}{dt},$$

so-called because its direction points along the curve at each t.

Definition 2.1.6 (Regular Parametrisation)**.** A parametrisation is regular if $\frac{d\boldsymbol{x}}{dt} \neq 0$ everywhere, and a curve is regular if it has at least one regular parametrisation.

Definition 2.1.7 (Singular Parametrisation and Cusps)**.** A parametrisation is singular if $\frac{d\boldsymbol{x}}{dt} = 0$ at at least one point. The singular points of a parametrisation are called susps.

Remark 2.1.8. You can construct a singular parametrisation even for a regular curve.

Remark 2.1.9. If we think of $\boldsymbol{x}(t)$ as the trajectory of a particle, then $\left|\frac{d\boldsymbol{x}}{dt}\right|$ is the speed at which the particle is moving, for each t.

Theorem 2.1.10 (Chain rule for parametrisation)**.** If $\boldsymbol{x}(t)$ and $\boldsymbol{x}(\tau)$ are two different parametrisations of the same curve, then the chain rule gives

$$\frac{d\boldsymbol{x}}{dt} = \frac{d\boldsymbol{x}}{d\tau}\frac{d\tau}{dt}.$$

2.2 Line Integral of Scalar Fields

Definition 2.2.1 (Line Integral of Scalar Field)**.** The line integral of a scalar field $f(\boldsymbol{x})$ along a curve C with parametrisation $\boldsymbol{x}(t)$, for $t \in [t_0, t_1]$, is defined as

$$\int_C f d\ell = \int_{t_0}^{t_1} f(\boldsymbol{x}(t)) \left|\frac{d\boldsymbol{x}}{dt}\right| dt$$

Proposition 2.2.2 (Formula for Scalar Line Element)**.** $\mathrm{d}\ell = \pm |\mathbf{x}'(t)|\, \mathrm{d}t$

Proposition 2.2.3 (Choice of Parametrisation does not matter)**.** The line integral of a scalar field is independent of the choice of parametrisation

Proof. If we to take two parametrisations of the same curve : $\mathbf{x}(t)$ for $t \in [t_0, t_1]$ and $\mathbf{x}(\tau)$ for $\tau \in [\tau_0, \tau_1]$, we can relate their tangent vectors by the chain rule:

$$\frac{d\boldsymbol{x}}{dt} = \frac{d\tau}{dt}\frac{d\boldsymbol{x}}{d\tau} \quad \Longrightarrow \left|\frac{d\boldsymbol{x}}{dt}\right| = \left|\frac{d\tau}{dt}\right| \left|\frac{d\boldsymbol{x}}{d\tau}\right|.$$

We now examine two cases, taking $\frac{d\mathbf{x}}{dt} > 0$ in both:

Case 1: $\mathbf{x}(\tau_0) = \mathbf{x}(t_0)$, meaning both parametrisations trace the curve in the same direction. Then:

$$\frac{d\tau}{dt} > 0 \Longrightarrow \int_{t_0}^{t_1} f(\boldsymbol{x}(t)) \left|\frac{d\boldsymbol{x}}{dt}\right| dt = \int_{t_0}^{t_1} f(\boldsymbol{x}(t)) \left|\frac{d\boldsymbol{x}}{d\tau}\right| \frac{d\tau}{dt} dt = \int_{\tau_0}^{\tau_1} f(\boldsymbol{x}(\tau)) \left|\frac{d\boldsymbol{x}}{d\tau}\right| d\tau.$$

Case 2: $\mathbf{x}(\tau_0) = \mathbf{x}(t_1)$, meaning the parametrisations trace the curve in opposite directions. Then:

$$\frac{d\tau}{dt} < 0 \quad \Longrightarrow \int_{t_0}^{t_1} f(\boldsymbol{x}(t)) \left|\frac{d\boldsymbol{x}}{dt}\right| dt = \int_{t_0}^{t_1} f(\boldsymbol{x}(t)) \left|\frac{d\boldsymbol{x}}{d\tau}\right| \left(-\frac{d\tau}{dt}\right) dt = -\int_{\tau_1}^{\tau_0} f(\boldsymbol{x}(\tau)) \left|\frac{d\boldsymbol{x}}{d\tau}\right| d\tau.$$

Once we negate the second integral, both formulas are recovered as identical.

□

Definition 2.2.4 (Rectifiable)**.** A curve which has finite length is called rectifiable. This is a weaker condition than "regular" (for example, if $\boldsymbol{x}(t)$ is Lipschitz continuous then it is rectifiable).

Remark 2.2.5 (Arclength)**.** We can derive the arc length between two points on the curve $\mathbf{r}(u)$, given by $u = u_1$ and $u = u_2$, as

$$s = \int_{u_1}^{u_2} \sqrt{\frac{d\mathbf{r}}{du} \cdot \frac{d\mathbf{r}}{du}} du \tag{2.1}$$

Proof. The tiny displacement vector is

$$d\mathbf{r} = \mathbf{r}'(u)du$$

Its magnitude gives the tiny distance travelled:

$$ds = |d\mathbf{r}|$$

Hence

$$ds = |\mathbf{r}'(u)du|$$

and since du is scalar,

$$ds = |\mathbf{r}'(u)|\,|du|$$

which becomes

$$ds = |\mathbf{r}'(u)|\,du$$

This can be directly integrated to find our arclength formula. □

2.3 Line Integrals of Vector Fields

Definition 2.3.1 (Unit Tangent Vector of Parametrisation)**.** The unit tangent vector of a parametrisation is

$$\hat{\boldsymbol{t}} = \frac{\frac{d\boldsymbol{x}}{dt}}{\left|\frac{d\boldsymbol{x}}{dt}\right|}.$$

This only depends on the direction of the parametrisation, not the "speed".

Definition 2.3.2 (Oriented Curve)**.** An oriented curve is a curve together with a specified (consistent) choice of $\hat{\mathbf{t}}$

Definition 2.3.3 (Line Integral of a Vector Field along Oriented Curve)**.** The line integral of a vector field $\boldsymbol{f}(\boldsymbol{x})$ along an oriented curve C with parametrisation $\boldsymbol{x}(t)$, for $t \in [t_0, t_1]$, is defined as the scalar line integral of the tangential component:

$$\int_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\ell = \int_{t_0}^{t_1} \boldsymbol{f}(\boldsymbol{x}(t)) \cdot \hat{\boldsymbol{t}} \left|\frac{d\boldsymbol{x}}{dt}\right| dt = \int_{t_0}^{t_1} \boldsymbol{f}(\boldsymbol{x}(t)) \cdot \frac{d\boldsymbol{x}}{dt} dt.$$

Example 2.3.4 (Showing Line Integral depends on Curve)**.** Take $\mathbf{F}(\mathbf{r}) = (xe^y, z^2, xy)$ and we want to find the line integral from $\mathbf{a} = (0,0,0)$ to $\mathbf{b} = (1,1,1)$.

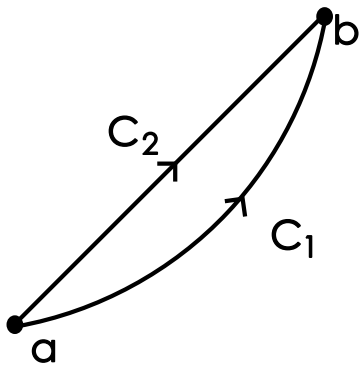


Figure 2.1: Two curves between a and b

We first integrate along the curve $C_1 : \mathbf{r}(u) = (u, u^2, u^3)$. Then $\mathbf{r}'(u) = (1, 2u, 3u^2)$, and $\mathbf{F}(\mathbf{r}(u)) = \left(ue^{u^2}, u^6, u^3\right)$. So

$$\begin{aligned}\int_{C_1} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} &= \int_0^1 \mathbf{F} \cdot \mathbf{r}'(u) \mathrm{d}u \\ &= \int_0^1 ue^{u^2} + 2u^7 + 3u^5 \, \mathrm{d}u \\ &= \frac{e}{2} - \frac{1}{2} + \frac{1}{4} + \frac{1}{2} \\ &= \frac{e}{2} + \frac{1}{4}\end{aligned}$$

Now we try to integrate along another curve $C_2 : \mathbf{r}(t) = (t, t, t)$. So $\mathbf{r}'(t) = (1, 1, 1)$.

$$\begin{aligned}\int_{C_2} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} &= \int \mathbf{F} \cdot \mathbf{r}'(t) \mathrm{d}t \\ &= \int_0^1 te^t + 2t^2 \, \mathrm{d}t \\ &= \frac{5}{3}\end{aligned}$$

We see that the line integral depends on the curve C in general, not just $\mathbf{a}, \mathbf{b}$.

Remark 2.3.5. The parametrisation must trace the curve in the direction given by $\hat{\mathbf{t}}$

Remark 2.3.6. The standard example from physics here is the work done by a particle.

Definition 2.3.7 (Circulation of $\mathbf{f}$ around C)**.** If the curve C is closed, this is often emphasised by the notation

$$\int_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\ell = \oint_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\ell$$

and the line integral is called the circulation of $\boldsymbol{f}$ around C.

Definition 2.3.8 (Piecewise Smooth Curve)**.** A piecewise smooth curve is a curve $C = C_1 + C_2 + \cdots + C_n$ with all C_i smooth with regular parametrisations. The line integral over a piecewise smooth C is

$$\int_C \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} = \int_{C_1} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} + \int_{C_2} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} + \cdots + \int_{C_n} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r}.$$

Example 2.3.9 (Turning two curves into piecewise smooth closed curve)**.** Take the example above, and let $C_3 = -C_2$. Then $C = C_1 + C_3$ is piecewise smooth but not smooth:

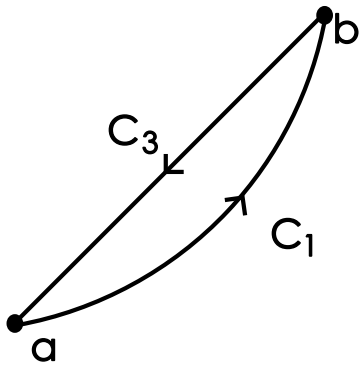


Figure 2.2: Closed line integral between a and b

Then

$$\begin{aligned}\oint_C \mathbf{F}\cdot\ \mathrm{d}\mathbf{r} &= \int_{C_1} \mathbf{F}\cdot\ \mathrm{d}\mathbf{r} + \int_{C_3} \mathbf{F}\cdot\ \mathrm{d}\mathbf{r} \\ &= \left(\frac{e}{2}+\frac{1}{4}\right) - \frac{5}{3} \\ &= -\frac{17}{12}+\frac{e}{2}\end{aligned}$$

Remark 2.3.10 (Differential form for Line Integrals)**.** You may also come across the "differential form" notation for line integrals,

$$\int_C \boldsymbol{f}\cdot\hat{\boldsymbol{t}}d\ell = \int_C \boldsymbol{f}\cdot d\boldsymbol{x}$$

2.4 Curvilinear Coordinates

Definition 2.4.1 (Curvilinear Coordinates)**.** In a curvilinear coordinate system (u, v, w), the coordinate lines - where two of the coordinates are constant and one varies - may be curved. These coordinate lines may be parametrised by the corresponding coordinate, so are given by $\boldsymbol{x}(u, v, w)$ with two of the coordinates fixed.

Definition 2.4.2 (Tangent Vectors and Unit Tangents)**.** For a system $\mathbf{x}(u, v, w)$ we have the corresponding tangent vectors:

$$\frac{\partial \boldsymbol{x}}{\partial u}, \frac{\partial \boldsymbol{x}}{\partial v}, \frac{\partial \boldsymbol{x}}{\partial w}$$

.

From these tangent vectors comes the unit tangent vectors:

$$\boldsymbol{e}_u = \frac{1}{h_u}\frac{\partial \boldsymbol{x}}{\partial u}, \quad \boldsymbol{e}_v = \frac{1}{h_v}\frac{\partial \boldsymbol{x}}{\partial v}, \quad \boldsymbol{e}_w = \frac{1}{h_w}\frac{\partial \boldsymbol{x}}{\partial w},$$

where

$$h_u = \left|\frac{\partial \boldsymbol{x}}{\partial u}\right|, \quad h_v = \left|\frac{\partial \boldsymbol{x}}{\partial v}\right|, \quad h_w = \left|\frac{\partial \boldsymbol{x}}{\partial w}\right|$$

are called the scale factors.

Remark 2.4.3. A set of tangent vectors form an orthonormal basis if the all of them are mutually orthogonal.

Remark 2.4.4. The scale factors tell us how much an infinitesimal region in (u, v, w) coordinates is stretched at different points in real space. They are crucial when calculating surface and volume integrals. We do not require systems to be orthogonal when we take their coordinates.

Definition 2.4.5 (Cylindrical and Spherical Coordinates)**.** Some alternative coordinate systems include

- **Cylindrical coordinates** are useful for problems with axial symmetry:

  $$\boldsymbol{x}(r, \theta, z) = r\cos\theta\,\boldsymbol{e}_1 + r\sin\theta\,\boldsymbol{e}_2 + z\,\boldsymbol{e}_3, \quad r > 0, \quad \theta \in [0, 2\pi], \quad z \in \mathbb{R},$$

  leading to scale factors $h_r = 1$, $h_\theta = r$, $h_z = 1$.
- **Spherical coordinates** are useful for problems with spherical symmetry:

  $$\begin{aligned}\boldsymbol{x}(r, \theta, \phi) = {} & r\sin\theta\cos\phi\,\boldsymbol{e}_1 + r\sin\theta\sin\phi\,\boldsymbol{e}_2 \\ & + r\cos\theta\,\boldsymbol{e}_3,\end{aligned}$$

  where $r > 0$, $\theta \in [0, \pi]$, $\phi \in [0, 2\pi]$, leading to scale factors $h_r = 1$, $h_\theta = r$, $h_\phi = r\sin\theta$.

Remark 2.4.6. It is relevant here to repeat the two coordinate basis from before:

Cylindrical polars	**Spherical polars**
Conversion formulae	
$x_1 = \rho\cos\varphi$ $x_2 = \rho\sin\varphi$ $x_3 = z$	$x_1 = r\sin\theta\cos\varphi$ $x_2 = r\sin\theta\sin\varphi$ $x_3 = r\cos\theta$
Basis vectors	
$\mathbf{e}_\rho = (\cos\varphi, \sin\varphi, 0)$ $\mathbf{e}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ $\mathbf{e}_z = (0, 0, 1)$	$\mathbf{e}_r = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$ $\mathbf{e}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ $\mathbf{e}_\theta = (\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta)$

Chapter 3

Surface and Volume Integrals

3.1 2D Surfaces

Definition 3.1.1 (2D Surface)**.** A 2-d surface embedded in $\mathbb{R}^3$ can be defined parametrically as a map $\boldsymbol{x}(u,v)$ from $\mathbb{R}^2 \to \mathbb{R}^3$, for $u \in [u_0, u_1]$ and $v \in [v_0, v_1]$. For each value of v, the position $\boldsymbol{x}(u,v)$ describes a curve on the surface with tangent vector $\frac{\partial \boldsymbol{x}}{\partial u}$. Similarly, for each value of u, we get a curve with tangent vector $\frac{\partial \boldsymbol{x}}{\partial v}$.

Definition 3.1.2 (Unit Normal to Surface)**.** If $(\,u, v\,)$ is a well-defined parametrisation, then their tangent vectors cannot be collinear. So the cross product of these tangent vectors allows us to define the unit normal to the surface:

$$\hat{\boldsymbol{n}} = \frac{\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}}{\left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right|}.$$

The sign of $\hat{\boldsymbol{n}}$ depends on the ordering of $(\,u, v\,)$. A surface together with a specified (consistent) choice of $\hat{\boldsymbol{n}}$ is called an oriented surface.

Remark 3.1.3. Not all surfaces are orientable, meaning that a consistent choice of $\hat{\mathbf{n}}$ exists.

Example 3.1.4 (Find the outward unit normal for the torus $\boldsymbol{x}(u,v) = (R{+}\cos v)\cos u\boldsymbol{e}_1{+}(R{+}\cos v$ sin u where $R > 1$ and $u, v \in [0, 2\pi)$. .]

The tangent vectors are

$$\frac{\partial \boldsymbol{x}}{\partial u} = -(R{+}\cos v)\sin u\boldsymbol{e}_1{+}(R{+}\cos v)\cos u\boldsymbol{e}_2, \quad \frac{\partial \boldsymbol{x}}{\partial v} = -\sin v\cos u\boldsymbol{e}_1{-}\sin v\sin u\boldsymbol{e}_2{+}\cos v\boldsymbol{e}_3,$$

so

$$\begin{aligned}\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v} &= \begin{vmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 \\ -(R+\cos v)\sin u & (R+\cos v)\cos u & 0 \\ -\sin v \cos u & -\sin v \sin u & \cos v \end{vmatrix} \\ &= (R+\cos v)\cos v \cos u \boldsymbol{e}_1 + (R+\cos v)\cos v \sin u \boldsymbol{e}_2 + (R+\cos v)\sin v \boldsymbol{e}_3 \\ &= (R+\cos v)\left(\cos v \cos u \boldsymbol{e}_1 + \cos v \sin u \boldsymbol{e}_2 + \sin v \boldsymbol{e}_3\right).\end{aligned}$$

This has magnitude

$$\left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| = |R+\cos v|\sqrt{\cos^2 v \cos^2 u + \cos^2 v \sin^2 u + \sin^2 v} = R + \cos v,$$

where the absolute value sign can be removed since $R > 1$. So the unit normal is

$$\hat{\boldsymbol{n}} = \cos v \cos u \boldsymbol{e}_1 + \cos v \sin u \boldsymbol{e}_2 + \sin v \boldsymbol{e}_3.$$

Definition 3.1.5 (Smooth and Piecewise Smooth)**.** A surface is called smooth if $\hat{\boldsymbol{n}}$ varies continuously over the surface (e.g. a sphere). It is called piecewise smooth if it can be divided into finitely many portions each of which is smooth (e.g. a cube).

Remark 3.1.6. It is important to learn to sketch the surfaces that you are asked about

Definition 3.1.7 (Open, Closed and Boundary)**.** A surface can be either **open** if it is bounded by a curve (e.g. the cone, or a hemisphere), or **closed** if it has no boundary (e.g. the sphere or torus). The **boundary** of an open surface S is a curve (or union of curves), often written ∂S. A simple type of surface is an **explicit surface** given by the height of a two-dimensional function, $z = h(x, y)$.

3.2 Surface Integrals of Scalar Fields

Definition 3.2.1 (Surface Integral of Scalar Field)**.** If $f(\boldsymbol{x}) = f(x, y, z)$ is a scalar field and S is a surface with parametrisation $\boldsymbol{x}(u, v)$, then we define the surface integral of f over S to be

$$\int_S f dS = \int_U f(\boldsymbol{x}(u,v)) \left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| du dv,$$

where U is the preimage of S in the (u, v) plane.

Remark 3.2.2. To motivate this formula, note that the area element $\left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| du dv$ is the area on the surface in $\mathbb{R}^3$ corresponding to an infinitesimal rectangle with sides (du, dv) in the (u, v) plane.

Proposition 3.2.3. If (u, v) are (two of a set of) orthogonal curvilinear coordinates, then

$$\left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| = h_u h_v.$$

Proof. Recall that $\frac{\partial \boldsymbol{x}}{\partial u} = h_u \boldsymbol{e}_u$ and $\frac{\partial \boldsymbol{x}}{\partial v} = h_v \boldsymbol{e}_v$. Since $h_u, h_v \geq 0$, we have that

$$\left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| = h_u h_v \left|\boldsymbol{e}_u \times \boldsymbol{e}_v\right| = h_u h_v \left|\pm \boldsymbol{e}_w\right| = h_u h_v.$$

□

Method 3.2.4 (Calculating Surface Area)**.** The surface area of S is computed by taking $f(\boldsymbol{x}) \equiv 1$, giving $|S| = \int_S dS$.

Example 3.2.5 (Use spherical coordinates to find the surface area of the sphere $x^2+y^2+z^2 = a^2$.)**.** The sphere radius a is given by $\boldsymbol{x}(\theta, \phi) = a\sin\theta\cos\phi \boldsymbol{e}_1 + a\sin\theta\sin\phi \boldsymbol{e}_2 + a\cos\theta \boldsymbol{e}_3$, and we know that the scale factors are $h_\theta = a$ and $h_\phi = a\sin\theta$. So from Proposition 2.1 we see that

$$\left|\frac{\partial \boldsymbol{x}}{\partial \theta} \times \frac{\partial \boldsymbol{x}}{\partial \phi}\right| = h_\theta h_\phi = a^2 \sin\theta.$$

The surface area is then

$$|S| = \int_S dS = \int_0^{2\pi} \left(\int_0^{\pi} a^2 \sin\theta d\theta \right) d\phi = a^2 \int_0^{2\pi} [-\cos\theta]_0^{\pi} d\phi = 2a^2 \int_0^{2\pi} d\phi = 4\pi a^2.$$

Proposition 3.2.6. The surface integral of a scalar field is independent of the choice of parametrisation.

Proof. Suppose $\boldsymbol{x}(u, v)$ and $\boldsymbol{x}(\mu, \nu)$ are two regular parametrisations of the same surface. By the chain rule,

$$\frac{\partial \boldsymbol{x}}{\partial u} = \frac{\partial \boldsymbol{x}}{\partial \mu}\frac{\partial \mu}{\partial u} + \frac{\partial \boldsymbol{x}}{\partial \nu}\frac{\partial \nu}{\partial u}, \quad \frac{\partial \boldsymbol{x}}{\partial v} = \frac{\partial \boldsymbol{x}}{\partial \mu}\frac{\partial \mu}{\partial v} + \frac{\partial \boldsymbol{x}}{\partial \nu}\frac{\partial \nu}{\partial v}.$$

So

$$\begin{aligned}\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v} &= \left(\frac{\partial \boldsymbol{x}}{\partial \mu}\frac{\partial \mu}{\partial u} + \frac{\partial \boldsymbol{x}}{\partial \nu}\frac{\partial \nu}{\partial u}\right) \times \left(\frac{\partial \boldsymbol{x}}{\partial \mu}\frac{\partial \mu}{\partial v} + \frac{\partial \boldsymbol{x}}{\partial \nu}\frac{\partial \nu}{\partial v}\right) \\ &= \left(\frac{\partial \boldsymbol{x}}{\partial \mu}\frac{\partial \mu}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial \nu}\frac{\partial \nu}{\partial v}\right) + \left(\frac{\partial \boldsymbol{x}}{\partial \nu}\frac{\partial \nu}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial \mu}\frac{\partial \mu}{\partial v}\right) \\ &= \left(\frac{\partial \mu}{\partial u}\frac{\partial \nu}{\partial v} - \frac{\partial \mu}{\partial v}\frac{\partial \nu}{\partial u}\right) \frac{\partial \boldsymbol{x}}{\partial \mu} \times \frac{\partial \boldsymbol{x}}{\partial \nu}.\end{aligned}$$

This means

$$\left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| = |J| \left|\frac{\partial \boldsymbol{x}}{\partial \mu} \times \frac{\partial \boldsymbol{x}}{\partial \nu}\right|,$$

where $J = \frac{\partial \mu}{\partial u}\frac{\partial \nu}{\partial v} - \frac{\partial \mu}{\partial v}\frac{\partial \nu}{\partial u}$ is the Jacobian from Calculus I for the change of variables from (μ, ν) to (u, v). Since $|J| du dv = d\mu d\nu$, we obtain

$$\int_U f(\boldsymbol{x}(u,v)) \left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| du dv = \int_{U'} f(\boldsymbol{x}(\mu,\nu)) \left|\frac{\partial \boldsymbol{x}}{\partial \mu} \times \frac{\partial \boldsymbol{x}}{\partial \nu}\right| d\mu d\nu,$$

where U' is the corresponding region of the (μ, ν) plane. □

Remark 3.2.7. If $S \subset \mathbb{R}^2$ is a flat surface in (say) the (x, y)-plane, then $\hat{\boldsymbol{n}}$ only has a z-component, so

$$\left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| = \left|\frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}\right| = |J|,$$

which is just the Jacobian for a change of coordinates from (x, y) to (u, v) in the plane.

3.3 Surface Integrals of Vector Fields

Definition 3.3.1 (Flux). If $\boldsymbol{f} : \mathbb{R}^3 \to \mathbb{R}^3$ is a vector field and S is an oriented surface with unit normal $\hat{\boldsymbol{n}}$, we define the surface integral (or flux) of $\boldsymbol{f}$ through S as

$$\int_S \boldsymbol{f} \cdot \hat{\boldsymbol{n}} dS.$$

Again, we evaluate the integral using a specific parametrisation $\boldsymbol{x}(u, v)$, so

$$\begin{aligned}\int_S \boldsymbol{f} \cdot \hat{\boldsymbol{n}} dS &= \int_U \boldsymbol{f}(\boldsymbol{x}(u,v)) \cdot \hat{\boldsymbol{n}} \left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| du dv \\ &= \int_U \boldsymbol{f}(\boldsymbol{x}(u,v)) \cdot \left(\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right) du dv\end{aligned}$$

where U is the preimage of S in the (u, v) plane.

Remark 3.3.2. A non-zero flux requires a component of $\boldsymbol{f}$ pointing "through" the surface. For given $|\boldsymbol{f}|$, the flux is largest if $\boldsymbol{f}$ is exactly perpendicular to the surface, but the flux also depends on the magnitude $|\boldsymbol{f}|$.

Definition 3.3.3 (Vector Area Element)**.** Sometimes you will see the shorthand notation $d\boldsymbol{S} = \hat{\boldsymbol{n}} dS$ for the "vector area element".

Remark 3.3.4. Unlike a scalar surface integral, a flux integral depends on the sign of $\hat{\boldsymbol{n}}$, i.e. on the orientation of the surface. You need to check that your chosen parametrisation $\boldsymbol{x}(u, v)$ gives the correct sign of $\hat{\boldsymbol{n}} = \frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}$. If not, interchange u and v.

Example 3.3.5 (Example 2.8: Calculate the outward flux of $\boldsymbol{f}(\boldsymbol{x}) = \cos^7 \boldsymbol{\theta} \boldsymbol{e}_r$ [in spherical coordinates)**.** through the hemisphere $r = a, z \geq 0$.]

Parametrising the sphere in spherical coordinates as $\boldsymbol{x}(\theta, \phi)$, recall that $\hat{\boldsymbol{n}} = \boldsymbol{e_r}$ and

$$\left| \frac{\partial \boldsymbol{x}}{\partial \theta} \times \frac{\partial \boldsymbol{x}}{\partial \phi} \right| = h_\theta h_\phi = a^2 \sin\theta.$$

Also note that the hemisphere is given by the half-range $\theta \in [0, \pi/2]$, with $\phi \in [0, 2\pi]$ as usual. So

$$\begin{aligned}
\int_S \boldsymbol{f} \cdot \hat{\boldsymbol{n}} dS &= \int_0^{2\pi} \left(\int_0^{\pi/2} \boldsymbol{f} \cdot \hat{\boldsymbol{n}} \left| \frac{\partial \boldsymbol{x}}{\partial \theta} \times \frac{\partial \boldsymbol{x}}{\partial \phi} \right| d\theta \right) d\phi \\
&= \int_0^{2\pi} \left(\int_0^{\pi/2} \left(\cos^7 \theta \boldsymbol{e}_r \right) \cdot \left(\boldsymbol{e}_r \right) a^2 \sin\theta d\theta \right) d\phi \\
&= a^2 \int_0^{2\pi} \left(\int_0^{\pi/2} \cos^7 \theta \sin\theta d\theta \right) d\phi \\
&= a^2 \int_0^{2\pi} \left[-\frac{1}{8} \cos^8 \theta \right]_0^{\pi/2} d\phi = \frac{a^2}{8} \int_0^{2\pi} d\phi = \frac{\pi a^2}{4}.
\end{aligned}$$

Definition 3.3.6 (Closed Surface Flux)**.** If the surface S is closed, this is often emphasised by writing

$$\int_S \boldsymbol{f} \cdot \hat{\boldsymbol{n}} dS = \oint_S \boldsymbol{f} \cdot \hat{\boldsymbol{n}} dS.$$

Definition 3.3.7 (Vector Valued Surface Integrals)**.** We will also meet vector-valued surface integrals of the form

$$\int_S \boldsymbol{f} dS$$

for some vector field $\boldsymbol{f} : \mathbb{R}^3 \to \mathbb{R}^3$.

3.4 Volume Integrals

Definition 3.4.1 (Volume Integral)**.** The volume integral of a scalar field $f : \mathbb{R}^3 \to \mathbb{R}$ over a 3 -d subregion $V \subset \mathbb{R}^3$ is

$$\int_V f dV = \int_U f(\boldsymbol{x}(u,v,w)) \left| \frac{\partial \boldsymbol{x}}{\partial u} \cdot \left(\frac{\partial \boldsymbol{x}}{\partial v} \times \frac{\partial \boldsymbol{x}}{\partial w} \right) \right| dudvdw,$$

where U is the preimage of V in (u,v,w) space, for some well-defined coordinates (u,v,w).

Remark 3.4.2. The factor $\left| \frac{\partial \boldsymbol{x}}{\partial u} \cdot \left(\frac{\partial \boldsymbol{x}}{\partial v} \times \frac{\partial \boldsymbol{x}}{\partial w} \right) \right| dudvdw$ is the volume of a parallelepiped in "real space" corresponding to an infinitesimal cuboid (du, dv, dw) in (u, v, w) space.

Proposition 3.4.3. If (u,v,w) are orthogonal curvilinear coordinates, then

$$\left| \frac{\partial \boldsymbol{x}}{\partial u} \cdot \left(\frac{\partial \boldsymbol{x}}{\partial v} \times \frac{\partial \boldsymbol{x}}{\partial w} \right) \right| = h_u h_v h_w.$$

Proof. Recall that $\frac{\partial \boldsymbol{x}}{\partial u} = h_u \boldsymbol{e}_u$, and similarly for the other two tangent vectors. Since $h_u, h_v, h_w \geq 0$,

$$\left| \frac{\partial \boldsymbol{x}}{\partial u} \cdot \left(\frac{\partial \boldsymbol{x}}{\partial v} \times \frac{\partial \boldsymbol{x}}{\partial w} \right) \right| = h_u h_v h_w \left| \boldsymbol{e}_u \cdot (\boldsymbol{e}_v \times \boldsymbol{e}_w) \right| = h_u h_v h_w \left| \boldsymbol{e}_u \cdot (\pm \boldsymbol{e}_u) \right| = h_u h_v h_w.$$

□

Remark 3.4.4. The factor $\left| \frac{\partial \boldsymbol{x}}{\partial u} \cdot \left(\frac{\partial \boldsymbol{x}}{\partial v} \times \frac{\partial \boldsymbol{x}}{\partial w} \right) \right|$ is often called the Jacobian (or Jacobian determinant). To see why, note that the scalar triple product may be written

$$\frac{\partial \boldsymbol{x}}{\partial u} \cdot \left(\frac{\partial \boldsymbol{x}}{\partial v} \times \frac{\partial \boldsymbol{x}}{\partial w} \right) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}.$$

Compare this to the 2-d case of a flat surface in the (x,y) plane, where we had

$$\left| \frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v} \right| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}.$$

Method 3.4.5 (Calculating Volume)**.** The volume of V is computed by taking $f(\boldsymbol{x}) \equiv$

1, giving $|V| = \int_V dV$.

Example 3.4.6 (Let V be the region bounded by the planes $x = 0, y = 0, z = 2$ and the surface $z = x^2 + y^2$ for $x, y \geq 0$. Calculate $\int_V x dV$.)**.** First we make a sketch to establish the shape of the region. Note that the surface $z = x^2 + y^2$ is a paraboloid.

Insert Sketch

Clearly this region has cylindrical symmetry, so the integral will be easiest to evaluate in cylindrical coordinates, $\boldsymbol{x}(r, \theta, z) = r\cos\theta \boldsymbol{e}_1 + r\sin\theta \boldsymbol{e}_2 + z\boldsymbol{e}_3$.

By Proposition 2.3 the Jacobian is $\left|\frac{\partial \boldsymbol{x}}{\partial r} \cdot \left(\frac{\partial \boldsymbol{x}}{\partial \theta} \times \frac{\partial \boldsymbol{x}}{\partial z}\right)\right| = h_r h_\theta h_z = r$. In our case, the boundaries $x = 0$ and $y = 0$ correspond to $\theta = 0$ and $\theta = \pi/2$, respectively. The range in r depends on the height, with maximum given by $z = x^2 + y^2 = r^2$, so that $r(z) \in [0, \sqrt{z}]$. Therefore we have

$$\begin{aligned}
\int_V x dV &= \int_0^2 \left(\int_0^{\pi/2} \left(\int_0^{\sqrt{z}} x r dr \right) d\theta \right) dz \\
&= \int_0^2 \left(\int_0^{\pi/2} \left(\int_0^{\sqrt{z}} r^2 \cos\theta dr \right) d\theta \right) dz \\
&= \int_0^2 \left(\int_0^{\pi/2} \cos\theta \left[\frac{r^3}{3} \right]_0^{\sqrt{z}} d\theta \right) dz \\
&= \int_0^2 \frac{z^{3/2}}{3} \left(\int_0^{\pi/2} \cos\theta d\theta \right) dz \\
&= \int_0^2 \frac{z^{3/2}}{3} [\sin\theta]_0^{\pi/2} dz = \int_0^2 \frac{z^{3/2}}{3} dz = \frac{2}{15} \left[z^{5/2} \right]_0^2 = \frac{2}{15} 2^{5/2} = \frac{8\sqrt{2}}{15}.
\end{aligned}$$

Proposition 3.4.7. The volume integral of a scalar field is independent of the choice of coordinates.

Proof. This is analogous to our proposition for a surface scalar integral , using the fact that a 3 -d change of variables obeys

$$dxdydz = \left| \frac{\partial \boldsymbol{x}}{\partial u} \cdot \left(\frac{\partial \boldsymbol{x}}{\partial v} \times \frac{\partial \boldsymbol{x}}{\partial w} \right) \right| dudvdw.$$

□

Remark 3.4.8. Occasionally you will come across a vector-valued volume integral, $\int_V f dV$. An example is the centre of mass of a solid body,

$$\overline{\boldsymbol{x}} = \frac{\int_V \rho(\boldsymbol{x}) \boldsymbol{x} dV}{\int_V \rho(\boldsymbol{x}) dV}$$

where $\rho(\boldsymbol{x})$ is the distribution of mass density within the body.

Chapter 4

Differential Operators

4.1 Differentiating Scalar Fields

Definition 4.1.1 (Gradient of a Scalar Field)**.** The gradient of a scalar field $f : \mathbb{R}^3 \to \mathbb{R}$ is the operator

$$\nabla f|_x = \lim_{\mathrm{d}(S)\to 0} \frac{1}{|V|} \oint_S f\hat{n} dS,$$

where S is a closed surface around $\boldsymbol{x}$ with outward normal $\hat{\boldsymbol{n}}$, enclosed volume $|V|$, and diameter $\mathrm{d}(S) = \sup_{\boldsymbol{x},\boldsymbol{y}\in S} |\boldsymbol{x} - \boldsymbol{y}|$.

Remark 4.1.2. Note that $\boldsymbol{\nabla} f$ is a vector field $\mathbb{R}^3 \to \mathbb{R}^3$. The intuition is that it points in the direction of fastest increase of f, and its magnitude tells you the maximum rate of increase.

Lemma 4.1.3. If a scalar field $f : \mathbb{R}^2 \to \mathbb{R}$ is continuous on a closed, bounded and connected domain $A \subset \mathbb{R}^2$ then there exists $\boldsymbol{x}_* \in A$ such that

$$\int_A f dS = |A| f(\boldsymbol{x}_*).$$

Main Idea / Strategy

- Having A be closed and bounded tells us it is compact. We can then guarantee a minimum and maximum for f on A
- We can produce an inequality bounding $f(\mathbf{x}$ and its minimum and maximum and then integrate the inequality
- By assuming the area ($|\ A\ |$) is none zero, we can divide by it to bound our integral

– Finally, the function is continuous and A is connected meaning $f(A)$ is connected and it follows that we have an average value.

Formal Proof

Since $A \subset \mathbb{R}^2$ is closed and bounded, it is compact. As f is continuous on A, the extreme value theorem implies that there exist $\boldsymbol{x}_m, \boldsymbol{x}_M \in A$ such that

$$m := f(\boldsymbol{x}_m) \leq f(\boldsymbol{x}) \leq f(\boldsymbol{x}_M) =: M$$

for all $\boldsymbol{x} \in A$. Integrating these inequalities over A, we obtain

$$m|A| \leq \int_A f dS \leq M|A|$$

Assuming $|A| > 0$, division by $|A|$ gives

$$m \leq \frac{1}{|A|} \int_A f dS \leq M$$

Now, since A is connected and f is continuous, the continuous image $f(A) \subset \mathbb{R}$ is connected. Every connected subset of $\mathbb{R}$ is an interval. Since $m, M \in f(A)$, it follows that

$$[m, M] \subseteq f(A).$$

Consequently, there exists $\boldsymbol{x}_* \in A$ such that

$$f(\boldsymbol{x}_*) = \frac{1}{|A|} \int_A f dS$$

Multiplying by $|A|$, we conclude that

$$\int_A f dS = |A| f(\boldsymbol{x}_*).$$

□

Proposition 4.1.4. In Cartesian coordinates,

$$\nabla f = \frac{\partial f}{\partial x}\boldsymbol{e}_1 + \frac{\partial f}{\partial y}\boldsymbol{e}_2 + \frac{\partial f}{\partial z}\boldsymbol{e}_3.$$

Main Idea / Strategy

Here we will use the given definition of the gradient and choose S to be the boundary of a small right circular cylinder centred at $\mathbf{x}(x, y, z)$ with its axis parallel to $\mathbf{e}_1$. We break this shape down:

- Evaluate the first component of the surface integral $\oint_S f\hat{\boldsymbol{n}}dS$. The curved surface contributes zero to this component because its outward normal has no $\boldsymbol{e}_1$ -component.
- The two circular end caps contribute the difference between the integrals of f over the two discs.
- Apply the previous integral mean value lemma to each disc integral. Their difference, divided by the cylinder's volume, becomes a difference quotient for f in the x -direction.
- Shrink the cylinder to $\boldsymbol{x}$. The difference quotient converges to $\partial f/\partial x$. Repeat for the other two components.

Formal Proof

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be continuously differentiable, and fix $\boldsymbol{x} = (x, y, z)$. We use the definition

$$\nabla f|_{\boldsymbol{x}} = \lim_{\mathrm{d}(S)\to 0} \frac{1}{|V|} \oint_S f\hat{\boldsymbol{n}}dS.$$

Take S to be the boundary of the cylinder

$$V_h = [x - h, x + h] \times D_r,$$

where D_r is the disc of radius r, centred at (y, z) in the yz-plane. Its volume is

$$|V_h| = 2h\pi r^2.$$

Write $S = S_+ \cup S_- \cup S_{\text{side}}$, where $S_+$and $S_-$are the end caps at $x + h$ and $x - h$, respectively.

On the curved side, the outward normal has zero $\boldsymbol{e}_1$-component. On the end caps, the outward normals are $\boldsymbol{e}_1$ and $-\boldsymbol{e}_1$, respectively. Hence

$$\begin{aligned}\oint_S f\hat{\boldsymbol{n}}\cdot\boldsymbol{e}_1 dS &= \int_{S_+} f dS - \int_S f dS \\ &= \int_{D_r} f\left(x+h,y',z'\right) dA \\ &\quad - \int_{D_r} f\left(x-h,y',z'\right) dA.\end{aligned}$$

Therefore, the first component of the gradient approximation is

$$\begin{aligned}G_1(h,r) = \frac{1}{2h\pi r^2}\Bigg[&\int_{D_r} f\left(x+h,y',z'\right) dA \\ &- \int_{D_r} f\left(x-h,y',z'\right) dA\Bigg].\end{aligned}$$

By the integral mean value lemma, applied to each disc, there exist points

$$\boldsymbol{p}_+ \in S_+, \quad \boldsymbol{p}_- \in S_-$$

such that

$$\int_{S_+} f dS = \pi r^2 f\left(\boldsymbol{p}_+\right), \quad \int_S f dS = \pi r^2 f\left(\boldsymbol{p}_-\right).$$

Consequently,

$$G_1(h,r) = \frac{f\left(\boldsymbol{p}_+\right) - f\left(\boldsymbol{p}_-\right)}{2h}.$$

Now choose a family of cylinders with $h \to 0$ and $r \to 0$, for example $r = h$. Then

$$\boldsymbol{p}_+ \to \boldsymbol{x}, \quad \boldsymbol{p}_- \to \boldsymbol{x},$$

but this alone does not establish that the quotient converges to the partial derivative: the two points may have different transverse coordinates. To justify the limit rigorously, we use the fundamental theorem of calculus on the original disc integrals instead:

$$\begin{aligned}G_1(h,r) &= \frac{1}{2h\pi r^2}\int_{D_r}\left[f\left(x+h,y',z'\right) - f\left(x-h,y',z'\right)\right] dA \\ &= \frac{1}{2h\pi r^2}\int_{D_r}\int_{x-h}^{x+h}\frac{\partial f}{\partial x}\left(s,y',z'\right) ds dA\end{aligned}$$

Thus $G_1(h,r)$ is the average value of $\partial f/\partial x$ over V_h. By continuity of $\partial f/\partial x$,

$$\lim_{h\to 0} G_1(h,h) = \frac{\partial f}{\partial x}(\boldsymbol{x}).$$

By choosing cylinders whose axes are parallel to $\boldsymbol{e}_2$ and $\boldsymbol{e}_3$, the same argument applies allowing us to define the gradient as the proposition suggests

□

Proposition 4.1.5. In orthogonal curvilinear coordinates (u, v, w), the gradient has the expression

$$\nabla f = \frac{1}{h_u}\frac{\partial f}{\partial u}\boldsymbol{e}_u + \frac{1}{h_v}\frac{\partial f}{\partial v}\boldsymbol{e}_v + \frac{1}{h_w}\frac{\partial f}{\partial w}\boldsymbol{e}_w.$$

Main Idea / Strategy

This directly follows on from the previous proposition by dot producting the new unit tangents with the cartesian formula.

Formal Proof

Looking at the first component

$$\boldsymbol{e}_u \cdot \nabla f = \left(\frac{1}{h_u}\frac{\partial \boldsymbol{x}}{\partial u}\right) \cdot \nabla f = \frac{1}{h_u}\left(\frac{\partial x}{\partial u}\frac{\partial f}{\partial x} + \frac{\partial y}{\partial u}\frac{\partial f}{\partial y} + \frac{\partial z}{\partial u}\frac{\partial f}{\partial z}\right) = \frac{1}{h_u}\frac{\partial f}{\partial u},$$

where the last step used the Chain Rule. The $\boldsymbol{e}_v$ and $\boldsymbol{e}_w$ components are analogous.

□

4.2 Differentiating Vector Fields

Definition 4.2.1 (Divergence and Curl of $\mathbf{f}$)**.** There are two ways to differentiate a vector field $\mathbf{f} : \mathbb{R}^3 \to \mathbb{R}^3$:

1. The divergence of $\boldsymbol{f}$ is the scalar field

$$\boldsymbol{\nabla} \cdot \boldsymbol{f}|_{\boldsymbol{x}} = \lim_{\mathrm{d}(S)\to 0} \frac{1}{|V|} \oint_S \hat{\boldsymbol{n}} \cdot \boldsymbol{f}\, dS.$$

2. The curl of $\boldsymbol{f}$ is the vector field

$$\boldsymbol{\nabla} \times \boldsymbol{f}|_{\boldsymbol{x}} = \lim_{\mathrm{d}(S)\to 0} \frac{1}{|V|} \oint_S \hat{\boldsymbol{n}} \times \boldsymbol{f} dS.$$

Remark 4.2.2. The intuition for divergence is that it measures whether (on average) the vector field is pointing outward ($\boldsymbol{\nabla} \cdot \boldsymbol{f} > 0$) or inward ($\boldsymbol{\nabla} \cdot \boldsymbol{f} < 0$) at a particular point.

PLACE IMAGE

The intuition for curl is that it measures how much (on average) the vector field at a particular point is rotating around any given axis.

PLACE IMAGE

Lemma 4.2.3. If $\boldsymbol{a} \in \mathbb{R}^3$ is a constant vector and $f : \mathbb{R}^3 \to \mathbb{R}$ is a C^1 scalar field, then

(i) $\boldsymbol{\nabla} \cdot (f\boldsymbol{a}) = (\boldsymbol{\nabla} f) \cdot \boldsymbol{a}$,

(ii) $\boldsymbol{\nabla} \times (f\boldsymbol{a}) = (\boldsymbol{\nabla} f) \times \boldsymbol{a}$.

Main Idea / Strategy

We just use the integral definitions of the formulas and then use the fact that $\mathbf{a}$ is constant to take it out.

Formal Proof

$$\boldsymbol{\nabla} \cdot (f\boldsymbol{a}) = \lim_{\mathrm{d}(S)\to 0} \frac{1}{|V|} \oint_S \hat{\boldsymbol{n}} \cdot (f\boldsymbol{a}) dS = \left(\lim_{\mathrm{d}(S)\to 0} \frac{1}{|V|} \oint_S f\hat{\boldsymbol{n}} dS \right) \cdot \boldsymbol{a} = (\boldsymbol{\nabla} f) \cdot \boldsymbol{a},$$

and

$$\boldsymbol{\nabla} \times (f\boldsymbol{a}) = \lim_{\mathrm{d}(S)\to 0} \frac{1}{|V|} \oint_S \hat{\boldsymbol{n}} \times (f\boldsymbol{a}) dS = \left(\lim_{\mathrm{d}(S)\to 0} \frac{1}{|V|} \oint_S f\hat{\boldsymbol{n}} dS \right) \times \boldsymbol{a} = (\boldsymbol{\nabla} f) \times \boldsymbol{a}.$$

□

Proposition 4.2.4. In Cartesian coordinates, divergence and curl are given by

(i) $\boldsymbol{\nabla} \cdot \boldsymbol{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$.

(ii)

$$\boldsymbol{\nabla}\times\boldsymbol{f} = \begin{vmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right)\boldsymbol{e}_1 + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x}\right)\boldsymbol{e}_2 + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right)\boldsymbol{e}_3.$$

Main Idea / Strategy

We use the lemma acting on constant vectors and consider the vector $(1,1,1)$ with respect to cartesian basis.

Formal Proof

$$\begin{aligned}\boldsymbol{\nabla}\cdot\boldsymbol{f} &= \boldsymbol{\nabla}\cdot(f_1e_1 + f_2e_2 + f_3e_3) \\ &= \boldsymbol{\nabla}\cdot(f_1e_1) + \boldsymbol{\nabla}\cdot(f_2e_2) + \boldsymbol{\nabla}\cdot(f_3e_3) \quad \text{[since the divergence is clearly linear]} \\ &= (\boldsymbol{\nabla}f_1)\cdot\boldsymbol{e}_1 + (\boldsymbol{\nabla}f_2)\cdot\boldsymbol{e}_2 + (\boldsymbol{\nabla}f_3)\cdot\boldsymbol{e}_3 \\ &= \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}. \quad \text{[from definition of the gradient]}\end{aligned}$$

(ii) Using Lemma 3.2 (ii), we have that

$$\boldsymbol{\nabla}\times(f_1\boldsymbol{e}_1) = (\boldsymbol{\nabla}f_1)\times\boldsymbol{e}_1 = \frac{\partial f_1}{\partial y}\boldsymbol{e}_2\times\boldsymbol{e}_1 + \frac{\partial f_1}{\partial z}\boldsymbol{e}_3\times\boldsymbol{e}_1 = -\frac{\partial f_1}{\partial y}\boldsymbol{e}_3 + \frac{\partial f_1}{\partial z}\boldsymbol{e}_2.$$

The other terms are obtained similarly.

□

Corollary 4.2.5. If $f:\mathbb{R}^3\to\mathbb{R}$ is a scalar field, and $\boldsymbol{f}:\mathbb{R}^3\to\mathbb{R}^3$ is a vector field, both with continuous second derivatives, then

(i) $\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{f}) \equiv 0$,

(ii) $\boldsymbol{\nabla}\times(\boldsymbol{\nabla}f) \equiv \mathbf{0}$.

Main Idea / Strategy

We simply use the Cartesian versions for divergence and curl since the expressions on the left are independent of any choice of coordinates. We then just apply the formulae

Formal Proof

For (i),

$$\begin{aligned}\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{f}) &= \frac{\partial}{\partial x}\left(\frac{\partial f_3}{\partial y}-\frac{\partial f_2}{\partial z}\right)+\frac{\partial}{\partial y}\left(\frac{\partial f_1}{\partial z}-\frac{\partial f_3}{\partial x}\right)+\frac{\partial}{\partial z}\left(\frac{\partial f_2}{\partial x}-\frac{\partial f_1}{\partial y}\right)\\ &= \frac{\partial^2 f_1}{\partial y\partial z}-\frac{\partial^2 f_1}{\partial z\partial y}+\frac{\partial^2 f_2}{\partial z\partial x}-\frac{\partial^2 f_2}{\partial x\partial z}+\frac{\partial^2 f_2}{\partial x\partial y}-\frac{\partial^2 f_3}{\partial y\partial x}=0\end{aligned}$$

where the mixed partial derivatives commute because they are continuous. For (ii),

$$\boldsymbol{\nabla}\times(\boldsymbol{\nabla}f)=\left(\frac{\partial^2 f}{\partial y\partial z}-\frac{\partial^2 f}{\partial z\partial y}\right)\boldsymbol{e}_1+\left(\frac{\partial^2 f}{\partial z\partial x}-\frac{\partial^2 f}{\partial x\partial z}\right)\boldsymbol{e}_2+\left(\frac{\partial^2 f}{\partial x\partial y}-\frac{\partial^2 f}{\partial y\partial x}\right)\boldsymbol{e}_3=\mathbf{0}.$$

□

4.3 Product Rules

Proposition 4.3.1 (Product Rules). Let ϕ,ψ be scalar fields and $\boldsymbol{f},\boldsymbol{g}$ be vector fields. Then

$$\begin{aligned}\nabla(\phi\psi) &= \phi\nabla\psi+\psi\nabla\phi,\\ \nabla\cdot(\phi\boldsymbol{f}) &= \phi(\nabla\cdot\boldsymbol{f})+\boldsymbol{f}\cdot\nabla\phi,\\ \nabla\times(\phi\boldsymbol{f}) &= \phi(\nabla\times\boldsymbol{f})+\nabla\phi\times\boldsymbol{f},\\ \nabla\cdot(\boldsymbol{f}\times\boldsymbol{g}) &= \boldsymbol{g}\cdot(\nabla\times\boldsymbol{f})-\boldsymbol{f}\cdot(\nabla\times\boldsymbol{g}),\\ \nabla\times(\boldsymbol{f}\times\boldsymbol{g}) &= \boldsymbol{f}(\nabla\cdot\boldsymbol{g})-\boldsymbol{g}(\nabla\cdot\boldsymbol{f})\\ &\quad+(\boldsymbol{g}\cdot\nabla)\boldsymbol{f}-(\boldsymbol{f}\cdot\nabla)\boldsymbol{g}.\end{aligned}$$

Main Idea / Strategy

Similarly, this can be proved true for all coordinate systems just using cartesian coordinates. We apply the operations carefully using the formulae.

Formal Proof

(i) We use the regular (single-variable) product rule to obtain

$$\begin{aligned}\boldsymbol{\nabla}(fg) &= \frac{\partial}{\partial x}(fg)\boldsymbol{e}_1 + \frac{\partial}{\partial y}(fg)\boldsymbol{e}_2 + \frac{\partial}{\partial z}(fg)\boldsymbol{e}_3 \\ &= \left(\frac{\partial f}{\partial x}g + f\frac{\partial g}{\partial x}\right)\boldsymbol{e}_1 + \left(\frac{\partial f}{\partial y}g + f\frac{\partial g}{\partial y}\right)\boldsymbol{e}_2 + \left(\frac{\partial f}{\partial z}g + f\frac{\partial g}{\partial z}\right)\boldsymbol{e}_3 \\ &= \left(\frac{\partial f}{\partial x}\boldsymbol{e}_1 + \frac{\partial f}{\partial y}\boldsymbol{e}_2 + \frac{\partial f}{\partial z}\boldsymbol{e}_3\right)g + f\left(\frac{\partial g}{\partial x}\boldsymbol{e}_1 + \frac{\partial g}{\partial y}\boldsymbol{e}_2 + \frac{\partial g}{\partial z}\boldsymbol{e}_3\right) \\ &= g\boldsymbol{\nabla}f + f\boldsymbol{\nabla}g.\end{aligned}$$

(ii) By Lemma 3.2 (i) we have

$$\begin{aligned}\boldsymbol{\nabla}\cdot(g\boldsymbol{f}) &= \boldsymbol{\nabla}\cdot(gf_1e_1 + gf_2e_2 + gf_3e_3) \\ &= \boldsymbol{\nabla}(gf_1)\cdot e_1 + \boldsymbol{\nabla}(gf_2)\cdot e_2 + \boldsymbol{\nabla}(gf_3)\cdot e_3 \\ &= f_1\boldsymbol{\nabla}g\cdot e_1 + g\boldsymbol{\nabla}f_1\cdot e_1 + f_2\boldsymbol{\nabla}g\cdot e_2 + g\boldsymbol{\nabla}f_2\cdot e_2 + f_3\boldsymbol{\nabla}g\cdot e_3 + g\boldsymbol{\nabla}f_3\cdot e_3 \\ &= (\boldsymbol{\nabla}g)\cdot(f_1e_1 + f_2e_2 + f_3e_3) + g\left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}\right) \\ &= (\boldsymbol{\nabla}g)\cdot\boldsymbol{f} + g\boldsymbol{\nabla}\cdot\boldsymbol{f}.\end{aligned}$$

(iii) Evaluating the cross product first gives

$$\begin{aligned}\boldsymbol{\nabla}\cdot(\boldsymbol{f}\times\boldsymbol{g}) =& \boldsymbol{\nabla}\cdot\begin{vmatrix} e_1 & e_2 & e_3 \\ f_1 & f_2 & f_3 \\ g_1 & g_2 & g_3 \end{vmatrix} \\ =& \frac{\partial}{\partial x}(f_2g_3 - f_3g_2) + \frac{\partial}{\partial y}(f_3g_1 - f_1g_3) + \frac{\partial}{\partial z}(f_1g_2 - f_2g_1) \\ =& g_1\left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right) + g_2\left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x}\right) + g_3\left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right) \\ &+ f_1\left(\frac{\partial g_2}{\partial z} - \frac{\partial g_3}{\partial y}\right) + f_2\left(\frac{\partial g_3}{\partial x} - \frac{\partial g_1}{\partial z}\right) + f_3\left(\frac{\partial g_1}{\partial y} - \frac{\partial g_2}{\partial x}\right) \\ =& \boldsymbol{g}\cdot(\boldsymbol{\nabla}\times\boldsymbol{f}) - \boldsymbol{f}\cdot(\boldsymbol{\nabla}\times\boldsymbol{g}).\end{aligned}$$

[Note the minus sign here, expected since $\boldsymbol{f}\times\boldsymbol{g}$ is antisymmetric.]

(iv) By Lemma 3.2 (ii), we have

$$\begin{aligned}
\boldsymbol{\nabla} \times (g\boldsymbol{f}) =& \boldsymbol{\nabla} \times (gf_1\boldsymbol{e}_1 + gf_2\boldsymbol{e}_2 + gf_3\boldsymbol{e}_3) \\
=& \boldsymbol{\nabla}(gf_1) \times \boldsymbol{e}_1 + \boldsymbol{\nabla}(gf_2) \times \boldsymbol{e}_2 + \boldsymbol{\nabla}(gf_3) \times \boldsymbol{e}_3 \\
=& \frac{\partial}{\partial y}(gf_1)\,\boldsymbol{e}_2 \times \boldsymbol{e}_1 + \frac{\partial}{\partial z}(gf_1)\,\boldsymbol{e}_3 \times \boldsymbol{e}_1 + \frac{\partial}{\partial x}(gf_2)\,\boldsymbol{e}_1 \times \boldsymbol{e}_2 + \frac{\partial}{\partial z}(gf_2)\,\boldsymbol{e}_3 \times \boldsymbol{e}_2 \\
& + \frac{\partial}{\partial x}(gf_3)\,\boldsymbol{e}_1 \times \boldsymbol{e}_3 + \frac{\partial}{\partial y}(gf_3)\,\boldsymbol{e}_2 \times \boldsymbol{e}_3 \\
=& -\frac{\partial}{\partial y}(gf_1)\,\boldsymbol{e}_3 + \frac{\partial}{\partial z}(gf_1)\,\boldsymbol{e}_2 + \frac{\partial}{\partial x}(gf_2)\,\boldsymbol{e}_3 - \frac{\partial}{\partial z}(gf_2)\,\boldsymbol{e}_1 - \frac{\partial}{\partial x}(gf_3)\,\boldsymbol{e}_2 + \frac{\partial}{\partial y}(gf_3)\,\boldsymbol{e}_1 \\
=& g\left[\left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right)\boldsymbol{e}_1 + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x}\right)\boldsymbol{e}_2 + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right)\boldsymbol{e}_3\right] \\
& \left(\frac{\partial g}{\partial y}f_3 - \frac{\partial g}{\partial z}f_2\right)\boldsymbol{e}_1 + \left(\frac{\partial g}{\partial z}f_1 - \frac{\partial g}{\partial x}f_3\right)\boldsymbol{e}_2 + \left(\frac{\partial g}{\partial x}f_2 - \frac{\partial g}{\partial y}f_1\right)\boldsymbol{e}_3 \\
=& g(\boldsymbol{\nabla} \times \boldsymbol{f}) + (\boldsymbol{\nabla} g) \times \boldsymbol{f}.
\end{aligned}$$

The rest are left till chapter 5 to reduce writing load

□

4.4 Divergence and Curl in Curvilinear Coordinates

Proposition 4.4.1. In orthogonal curvilinear coordinates (u, v, w), the divergence and curl are given by

(i) $\boldsymbol{\nabla} \cdot \boldsymbol{f} = \frac{1}{h_u h_v h_w}\left[\frac{\partial}{\partial u}(h_v h_w f_u) + \frac{\partial}{\partial v}(h_u h_w f_v) + \frac{\partial}{\partial w}(h_u h_v f_w)\right]$.

(ii) $\boldsymbol{\nabla} \times \boldsymbol{f} = \frac{1}{h_u h_v h_w}\begin{vmatrix} h_u\boldsymbol{e}_u & h_v\boldsymbol{e}_v & h_w\boldsymbol{e}_w \\ \frac{\partial}{\partial u} & \frac{\partial}{\partial v} & \frac{\partial}{\partial w} \\ h_u f_u & h_v f_v & h_w f_w \end{vmatrix}$.

Main Idea / Strategy

The coordinate system is orthogonal and so we can rewrite any unit tangent in terms of the others. We then use the formula for the gradient in curvilinear coordinates to rewrite aspects of the expanded equation. We can then show that each component follows similarly:

Formal Proof

First, note from Proposition 3.2 that $\boldsymbol{\nabla} v = \frac{1}{h_v}\boldsymbol{e}_v$, so $\boldsymbol{e}_v = h_v \boldsymbol{\nabla} v$, and similarly $\boldsymbol{e}_w = h_w \boldsymbol{\nabla} w$.

For (i), note that using orthogonality followed by Proposition 3.4 (ii), we get

$$
\begin{aligned}
\boldsymbol{\nabla}\cdot(f_u\boldsymbol{e}_u) &= \boldsymbol{\nabla}\cdot(f_u\boldsymbol{e}_v\times\boldsymbol{e}_w)\\
&= \boldsymbol{\nabla}\cdot(f_u h_v h_w\boldsymbol{\nabla}v\times\boldsymbol{\nabla}w)\\
&= \boldsymbol{\nabla}(f_u h_v h_w)\cdot\boldsymbol{\nabla}v\times\boldsymbol{\nabla}w + f_u h_v h_w\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}v\times\boldsymbol{\nabla}w)\\
&= \boldsymbol{\nabla}(f_u h_v h_w)\cdot\boldsymbol{\nabla}v\times\boldsymbol{\nabla}w + f_u h_v h_w\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times[v\boldsymbol{\nabla}w]) \quad \text{[from previous lecture]}\\
&= \boldsymbol{\nabla}(f_u h_v h_w)\cdot\boldsymbol{\nabla}v\times\boldsymbol{\nabla}w \quad [\text{ using } \boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{f})\equiv 0]\\
&= \boldsymbol{\nabla}(f_u h_v h_w)\cdot\frac{\boldsymbol{e}_v\times\boldsymbol{e}_w}{h_v h_w}\\
&= \frac{1}{h_v h_w}\boldsymbol{e}_u\cdot\boldsymbol{\nabla}(f_u h_v h_w)\\
&= \frac{1}{h_u h_v h_w}\frac{\partial}{\partial u}(h_v h_w f_u),
\end{aligned}
$$

where the last step used Proposition 3.2. The other components are similar. For (ii), we use $\boldsymbol{e}_u = h_u\boldsymbol{\nabla}u$ and Proposition 3.4 (iv) to find

$$
\begin{aligned}
\boldsymbol{\nabla}\times(f_u\boldsymbol{e}_u) &= \boldsymbol{\nabla}\times(f_u h_u\boldsymbol{\nabla}u)\\
&= \boldsymbol{\nabla}(f_u h_u)\times\boldsymbol{\nabla}u + f_u h_u\boldsymbol{\nabla}\times\boldsymbol{\nabla}u\\
&= \boldsymbol{\nabla}(f_u h_u)\times\boldsymbol{\nabla}u \quad [\text{ using } \boldsymbol{\nabla}\times\boldsymbol{\nabla}f = 0]\\
&= \frac{1}{h_u}\boldsymbol{\nabla}(f_u h_u)\times\boldsymbol{e}_u\\
&= \frac{1}{h_u}\boldsymbol{\nabla}(f_u h_u)\times(\boldsymbol{e}_v\times\boldsymbol{e}_w)\\
&= \frac{1}{h_u}\boldsymbol{e}_v[\boldsymbol{e}_w\cdot\boldsymbol{\nabla}(f_u h_u)] - \boldsymbol{e}_w[\boldsymbol{e}_v\cdot\boldsymbol{\nabla}(f_u h_u)]\\
&= \frac{1}{h_u h_w}\frac{\partial}{\partial w}(h_u f_u)\boldsymbol{e}_v - \frac{1}{h_u h_v}\frac{\partial}{\partial v}(h_u f_u)\boldsymbol{e}_w.
\end{aligned}
$$

The penultimate step used the vector triple product identity $\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a}\cdot\boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a}\cdot\boldsymbol{b})$. Again, the other components are similar.

□

Chapter 5

Integral Theorems

5.1 Fundamental Theorem of Line Integrals

Theorem 5.1.1 (Fundamental Theorem of Calculus)**.** If $f'(x)$ is continuous on $[a, b]$ then

$$\int_a^b f'(t)dt = f(b) - f(a).$$

Theorem 5.1.2 (Fundamental Theorem of Line Integrals)**.** If $f : \mathbb{R}^n \to \mathbb{R}$ is a C^1 scalar field and C is an oriented curve with parametrisation $\boldsymbol{x}(t)$ for $t \in [t_0, t_1]$, then

$$\int_C \nabla f \cdot \hat{\boldsymbol{t}} d\ell = f\left(\boldsymbol{x}\left(t_1\right)\right) - f\left(\boldsymbol{x}\left(t_0\right)\right).$$

Main Idea / Strategy

Formal Proof

□

Definition 5.1.3 (Conservative Field in a Potential)**.** A vector field $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^n$ is conservative if it can be written as $\boldsymbol{f} = \boldsymbol{\nabla} g$ for some C^1 scalar field $g : \mathbb{R}^n \to \mathbb{R}$ called a (scalar) potential.

Proposition 5.1.4 (Path Independence)**.** A vector field is conservative if and only if it has path independent line integrals between every pair of points.

Main Idea / Strategy

Formal Proof

□

5.2 Divergence Theorem

Theorem 5.2.1 (Divergence Theorem)**.** If $\boldsymbol{f} : \mathbb{R}^3 \to \mathbb{R}^3$ is a C^1 vector field and S is a closed surface with outward normal $\hat{\boldsymbol{n}}$ bounding a region V, then

$$\oint_S \boldsymbol{f} \cdot \hat{\boldsymbol{n}} dS = \int_V (\boldsymbol{\nabla} \cdot \boldsymbol{f}) dV.$$

Remark 5.2.2. The total amount of a vector field flowing out through a closed surface equals the total divergence of the field throughout the volume enclosed by that surface.

Main Idea / Strategy

Formal Proof

□

5.3 Stokes' Theoren

Theorem 5.3.1 (Stokes' Theorwm)**.** If $\boldsymbol{f} : \mathbb{R}^3 \to \mathbb{R}^3$ is a C^1 vector field and C is an oriented closed curve bounding a surface S, then

$$\oint_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\boldsymbol{\ell} = \int_S (\boldsymbol{\nabla} \times \boldsymbol{f}) \cdot \hat{\boldsymbol{n}} dS$$

where $\hat{\boldsymbol{n}}$ is chosen such that $\hat{\boldsymbol{t}} \times \hat{\boldsymbol{n}}$ on C is outward.

Theorem 5.3.2 (Green's Theorem)**.** If $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^2$ is a C^1 vector field and C is a closed curve in $\mathbb{R}^2$ oriented anti-clockwise bounding a region A, then

$$\oint_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\ell = \int_A \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) dxdy.$$

Corollary 5.3.3 (Conservative Fields have zero curl)**.** If $\boldsymbol{f} : \mathbb{R}^3 \to \mathbb{R}^3$ is a C^1 vector field with $\boldsymbol{\nabla} \times \boldsymbol{f} = \boldsymbol{0}$ throughout a simply connected region $V \subset \mathbb{R}^3$, then $\boldsymbol{f}$ is conservative in V.

Main Idea / Strategy

Formal Proof

□

Definition 5.3.4 (Simply Connected)**.** A region $V \subset \mathbb{R}^3$ is simply connected if any closed curve in V can be continuously shrunk to a point in V. A ball (the interior of a sphere) is simply connected, but a toroid (interior of a torus) is not:

5.4 Proof of Stokes' Theorem

Lemma 5.4.1. Let C be a curve in the plane having unit normal $\hat{\boldsymbol{n}}$, with diameter $\mathrm{d}(C)$, enclosed area $|A|$, and containing the point $\boldsymbol{x}$. For any C^1 vector field $\boldsymbol{f} : \mathbb{R}^3 \to \mathbb{R}^3$,

$$\hat{\boldsymbol{n}} \cdot (\boldsymbol{\nabla} \times \boldsymbol{f})|_{\boldsymbol{x}} = \lim_{\mathrm{d}(C)\to 0} \frac{1}{|A|} \oint_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\ell,$$

where C is parametrised such that $\hat{\boldsymbol{t}} = \hat{\boldsymbol{n}} \times \hat{\boldsymbol{n}}_c$, with $\hat{\boldsymbol{n}}_c$ being the outward normal to the curve.

Main Idea / Strategy

Formal Proof

□

Stokes' Theorem Proof

Main Idea / Strategy

Formal Proof

□

Chapter 6

Index Notation

6.1 Scalar Products

Definition 6.1.1 (Index Notation)**.** *A* vector equation such as

$$\boldsymbol{c} = \boldsymbol{a} + \boldsymbol{b}$$

is written in index notation as

$$c_i = a_i + b_i,$$

where it is understood that the equation holds for $i = 1, \ldots, n$, meaning an arbitrary component in the standard (Cartesian) basis $\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n\}$.

Remark 6.1.2. An index that appears once in each term is called a free index. The choice of letter is arbitrary, but all terms in an equation must have the same free index (or indices).

Definition 6.1.3 (Einstein Summation Convention)**.** In index notation, we write the scalar product as

$$\boldsymbol{a} \cdot \boldsymbol{b} = a_j b_j,$$

where the repeated index indicates that the term should be summed from $j = 1$ to n. This implied summation is called the **(Einstein) summation convention**.

An index that appears exactly twice in a term is called a **dummy index**, and always implies a summation over that index. Again, the choice of letter is arbitrary.

Definition 6.1.4 (Kronecker Delta)**.**

$$\boldsymbol{a}\cdot\boldsymbol{b}=(a_1,a_2,a_3)\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}b_1\\b_2\\b_3\end{pmatrix}=a_1b_1+a_2b_2+a_3b_3.$$

In index notation, this may be written

$$\boldsymbol{a}\cdot\boldsymbol{b}=\delta_{ij}a_ib_j,$$

where the components of the identity matrix are given by the Kronecker delta, defined as

$$\delta_{ij}=\begin{cases}1 & \text{if } i=j,\\ 0 & \text{if } i\neq j.\end{cases}$$

Proposition 6.1.5. Multiplying an expression with free index i by δ_{ij} is equivalent to replacing $i\to j$. We can simplify expressions by replacing the Kronecker Delta with a replacement instead.

Main Idea / Strategy

Formal Proof

□

6.2 Vector Products

Definition 6.2.1 (Alternating Tensor (Levi-Civita) Symbol)**.** In $\mathbb{R}^3$, we can write vector products in index notation by introducing the alternating tensor (or Levi-Civita symbol),

$$\epsilon_{ijk}=\begin{cases}0 & \text{if any of } i,j,k \text{ are equal,}\\ +1 & \text{if } (i,j,k)=(1,2,3),(2,3,1) \text{ or } (3,1,2),\\ -1 & \text{if } (i,j,k)=(1,3,2),(2,1,3) \text{ or } (3,2,1).\end{cases}$$

Remark 6.2.2. This allows us to express the vector product in index notation as

$$[\boldsymbol{a}\times\boldsymbol{b}]_i=\epsilon_{ijk}a_jb_k.$$

Example 6.2.3. Note that this scalar triple product is a scalar, so there should be no free index. We have

$$\begin{aligned}
\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c}) &= a_i[\boldsymbol{b}\times\boldsymbol{c}]_i \\
&= a_i\epsilon_{ijk}b_jc_k \\
&= \epsilon_{ijk}a_ib_jc_k \quad [\text{ order doesn't matter for scalars }] \\
&= \epsilon_{kij}a_ib_jc_k \\
&= c_k\epsilon_{kij}a_ib_j \\
&= c_i\epsilon_{ijk}a_jb_k \quad [\text{ relabelling the dummy indices } (i,j,k)\to(j,k,i)] \\
&= c_i[\boldsymbol{a}\times\boldsymbol{b}]_i \\
&= \boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b}).
\end{aligned}$$

The final relabelling wasn't strictly necessary, just aesthetically pleasing!

Remark 6.2.4. You can relabel dummy indicies as needed to aid you.

Proposition 6.2.5 (The $\epsilon-\delta$ identity)**.**

$$\epsilon_{ijk}\epsilon_{klm}=\delta_{il}\delta_{jm}-\delta_{im}\delta_{jl}.$$

Main Idea / Strategy

Formal Proof

□

6.3 Derivatives

Definition 6.3.1 (Grad, Div and Curl in Index Notation)**.**

$$\boldsymbol{\nabla}f=\frac{\partial f}{\partial x}\boldsymbol{e}_1+\frac{\partial f}{\partial y}\boldsymbol{e}_2+\frac{\partial f}{\partial z}\boldsymbol{e}_3,\quad \boldsymbol{\nabla}\cdot\boldsymbol{f}=\frac{\partial f_1}{\partial x}+\frac{\partial f_2}{\partial y}+\frac{\partial f_3}{\partial z},\quad \boldsymbol{\nabla}\times\boldsymbol{f}=\begin{vmatrix}\boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ f_1 & f_2 & f_3\end{vmatrix}$$

Therefore the gradient, divergence and curl are written in index notation as

$$[\nabla f]_i=\frac{\partial f}{\partial x_i},\quad \nabla\cdot f=\frac{\partial f_j}{\partial x_j},\quad [\nabla\times f]_i=\epsilon_{ijk}\frac{\partial f_k}{\partial x_j}.$$

Remark 6.3.2. We can complete the product rule proof here using this notation

The i component of the left-hand side is

$$
\begin{aligned}
[\boldsymbol{\nabla} \times (\boldsymbol{f} \times \boldsymbol{g})]_i &= \epsilon_{ijk} \frac{\partial}{\partial x_j} (\epsilon_{klm} f_l g_m) \\
&= \epsilon_{ijk}\epsilon_{klm} \frac{\partial}{\partial x_j} (f_l g_m) \\
&= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) \frac{\partial}{\partial x_j} (f_l g_m) \quad [\text{ by Proposition 5.2 }] \\
&= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) \left(\frac{\partial f_l}{\partial x_j} g_m + f_l \frac{\partial g_m}{\partial x_j} \right) \\
&= \delta_{il}\delta_{jm} \left(\frac{\partial f_l}{\partial x_j} g_m + f_l \frac{\partial g_m}{\partial x_j} \right) - \delta_{im}\delta_{jl} \left(\frac{\partial f_l}{\partial x_j} g_m + f_l \frac{\partial g_m}{\partial x_j} \right) \\
&= \frac{\partial f_i}{\partial x_j} g_j + f_i \frac{\partial g_j}{\partial x_j} - \frac{\partial f_j}{\partial x_j} g_i - f_j \frac{\partial g_i}{\partial x_j} \quad [\text{ by Proposition 5.1 }] \\
&= \frac{\partial f_i}{\partial x_j} g_j + f_i (\boldsymbol{\nabla} \cdot \boldsymbol{g}) - (\boldsymbol{\nabla} \cdot \boldsymbol{f}) g_i - f_j \frac{\partial g_i}{\partial x_j}.
\end{aligned}
$$

This time the right-hand side contains "triple products", so we start with that and simplify. We have

$$
\begin{aligned}
[\boldsymbol{f} \times (\boldsymbol{\nabla} \times \boldsymbol{g})]_i &= \epsilon_{ijk} f_j \epsilon_{klm} \frac{\partial g_m}{\partial x_l} \\
&= \epsilon_{ijk}\epsilon_{klm} f_j \frac{\partial g_m}{\partial x_l} \\
&= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) f_j \frac{\partial g_m}{\partial x_l} \\
&= \delta_{il}\delta_{jm} f_j \frac{\partial g_m}{\partial x_l} - \delta_{im}\delta_{jl} f_j \frac{\partial g_m}{\partial x_l} \quad [\text{by Proposition 5.2}] \\
&= f_j \frac{\partial g_j}{\partial x_i} - f_j \frac{\partial g_i}{\partial x_j} \quad [\text{by Proposition 5.1}].
\end{aligned}
$$

So the i component of the full right-hand side of (vi) is

$$
\begin{aligned}
[\text{RHS}_i &= g_j \frac{\partial f_i}{\partial x_j} + f_j \frac{\partial g_i}{\partial x_j} + f_j \frac{\partial g_j}{\partial x_i} - f_j \frac{\partial g_i}{\partial x_j} + g_j \frac{\partial f_j}{\partial x_i} - g_j \frac{\partial f_i}{\partial x_j} \\
&= f_j \frac{\partial g_j}{\partial x_i} + g_j \frac{\partial f_j}{\partial x_i} \\
&= \frac{\partial}{\partial x_i} (f_j g_j) = [\boldsymbol{\nabla} (\boldsymbol{f} \cdot \boldsymbol{g})]_i.
\end{aligned}
$$

6.4 Second Derivatives

Definition 6.4.1 (Second Derivatives). For a scalar field $g : \mathbb{R}^3 \to \mathbb{R}$ or a vector field $\boldsymbol{f} : \mathbb{R}^3 \to \mathbb{R}^3$, there are five ways to combine two derivatives to get a second derivative operator:

$$\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}g),\quad \boldsymbol{\nabla}\times(\boldsymbol{\nabla}g),\quad \boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{f}),\quad \boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{f}),\quad \text{or}\quad \boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{f}).$$

However, we already know that $\boldsymbol{\nabla}\times(\boldsymbol{\nabla}g)=\mathbf{0}$ and $\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{f})=0$, so there are only three non-trivial operators:

- The Laplacian of a scalar field is the operator defined by

$$\nabla^2 g=\nabla\cdot(\nabla g).$$

- Next, consider the operator $\boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{f})$. Its i component is simply

$$[\boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{f})]_i=\frac{\partial}{\partial x_i}\left(\frac{\partial f_j}{\partial x_j}\right).$$

- Finally we consider:

$$\nabla\times(\nabla\times f)=\nabla(\nabla\cdot f)-\nabla^2 f.$$

Proposition 6.4.2. In orthogonal curvilinear coordinates (u,v,w), the Laplacian is given by

$$\nabla^2 g=\frac{1}{h_u h_v h_w}\left[\frac{\partial}{\partial u}\left(\frac{h_v h_w}{h_u}\frac{\partial g}{\partial u}\right)+\frac{\partial}{\partial v}\left(\frac{h_u h_w}{h_v}\frac{\partial g}{\partial v}\right)+\frac{\partial}{\partial w}\left(\frac{h_u h_v}{h_w}\frac{\partial g}{\partial w}\right)\right].$$

Main Idea / Strategy

Formal Proof

□

Remark 6.4.3. Be very careful with $\boldsymbol{\nabla}^2\boldsymbol{f}$ in non-Cartesian coordinate systems! For example, in orthogonal curvilinear coordinates, $[\nabla^2 f]\cdot e_u\neq\nabla^2 f_u$. To find the correct expression, you would rearrange and calculate

$$\nabla^2 f=\nabla(\nabla\cdot f)-\nabla\times(\nabla\times f).$$

This shows, incidentally, that $\boldsymbol{\nabla}^2\boldsymbol{f}$ really is coordinate independent, since everything on the righthand side is.

Part II

Partial Differential Equations

Chapter 7

Mode Decomposition of Functions

7.1 Finite vs Infinite dimensional vector spaces

A common technique in mathematical analysis is to decompose an object into simple, independent pieces in order to determine their properties. This idea is central to analysis as a whole; examples include Taylor's series and Fourier series.

Definition 7.1.1 (Modes and Mode Decomposition)**.** If

$$f(x) = \sum_n c_n \psi_n(x),$$

we will often refer to the basis functions ψ_n as the **modes** of the decomposition.

A mode decomposition is therefore breaking down a complicated object into the sum of similar compnents whose amplitudes can be studied separately.

Later, when solving PDEs, the natural modes will be selected by a differential operator together with its boundary conditions. With vectors we can, for a given $\boldsymbol{a}$, find its kth component using the coefficient grab method:

$$\boldsymbol{a} \cdot \boldsymbol{e_k} = (a_1 \boldsymbol{e}_1 + a_2 \boldsymbol{e}_2 + \ldots a_n \boldsymbol{e}_n) \cdot \boldsymbol{e_k} = a_k \boldsymbol{e}_k \cdot \boldsymbol{e}_k,$$

where we have used orthogonality, $\boldsymbol{e}_i \cdot \boldsymbol{e}_j = 0$ unless $i = j$, for the last step. Then we have the coefficient formula:

$$a_k = \frac{a \cdot e_k}{e_k \cdot e_k}.$$

This formula is used to obtain the set $\{a_k\}_{k=1}^n$. Our goal is to generalise this idea so that we can use it in PDE analysis.

7.2 Function Spaces

This decomposition is comparable to vector spaces seen in Linear Algebra. Our goal is to determine some basis of functions that can be used to define a function space. From here, we will define a unique function as a summation of these basis functions together with a set of coefficients.

An example where we have seen this is the Fourier series:

$$f(x) = a_0 + \sum_{n=1}^{\infty}[a_n + \cos(\frac{n\pi x}{L}) + b_n \sin(\frac{n\pi x}{L})].$$

Much like the vector space equivalent, we can make the coefficients depend on another parameter, for instance time:

$$f(x) = a_0 + \sum_{n=1}^{\infty}[a_n(t) + \cos(\frac{n\pi x}{L}) + b_n(t) \sin(\frac{n\pi x}{L})].$$

So since we have reached a parallel to the vector space description, we must extend some of our objects from Linear Algebra to their function equivalent.

Definition 7.2.1 (Inner product of functions)**.** The inner product of a pair of functions $u(x), v(x)$ which are suitably well behaved on the domain $x \in [a, b]$ is defined to be the following integral product:

$$\langle u, v\rangle = \int_a^b u(x)\overline{v(x)}\mathrm{d}x.$$

And also,

Definition 7.2.2 (Orthogonal Functions)**.** A pair of functions $u(x), v(x)$ are orthogonal on $x \in [a, b]$ with weight $r(x)$, if their weighted inner product is zero:

$$\langle r(x)u, v\rangle = \langle u, r(x)v\rangle = \int_a^b r(x)u(x)v(x)\mathrm{d}x = 0.$$

which of course is the equivalent of vector orthogonality $\langle \boldsymbol{u}, \boldsymbol{v}\rangle = \boldsymbol{u} \cdot \boldsymbol{v} = 0$.

Remark 7.2.3 (Ensuring $r(x)$ is a positive function generalises to the norm)**.** It is important that this weighting is a positive function on the domain interior. This ensures that the inner product of a function with itself is positive so we can define the notion of the "length" of a function:

$$\langle u, r(x)u\rangle = \|u\|^2 > 0, \text{ unless } u = 0.$$

This is the weighted norm of the function $\boldsymbol{u}$. If this notion (a norm) is applied to vectors $\langle \boldsymbol{u}, \boldsymbol{u} \rangle = \boldsymbol{u} \cdot \boldsymbol{u}$ we usually refer to it as the length of the vector. The term norm is the generalisation.

7.2.1 The general coefficient grab

Method 7.2.4. Let's assume we have some non-zero family of functions $\psi_n(x) n = 0, 1, 2, \ldots \infty$, and that they are orthogonal with some weight function $r(x)$ on some domain $x \in [a, b]$:

$$\langle \psi_n, r(x)\psi_m \rangle = 0, \text{ if } n \neq m.$$

We want to describe some arbitrary function $f(x)$ on $x \in [a, b]$, and assume its weighted norm is finite, $\langle f, r(x)f \rangle < \infty$. This is the weighted square-integrable space often denoted $L^2_r([a, b])$. Our aim is then to find coefficients a_n such that:

$$f(x) = \sum_{n=0}^{\infty} a_n \psi_n(x).$$

To get the coefficient a_m we use the coefficient grab method. We first multiply by the basis function $\psi_m(x)$ and weight $r(x)$, then apply the inner product (integrate):

$$f(x)r(x)\psi_m(x) = \sum_{n=0}^{\infty} a_n \psi_n(x) r(x) \psi_m(x),$$
$$\langle f(x), r(x)\psi_m \rangle = \sum_{n=0}^{\infty} a_n \langle \psi_n(x), r(x)\psi_m \rangle = a_m \langle \psi_m(x), r(x)\psi_m \rangle ,$$

where we have imposed linearity of the inner product and orthogonality. Then we have the coefficient formula:

$$a_m = \frac{\langle f(x), r(x)\psi_m \rangle}{\langle \psi_m(x), r(x)\psi_m \rangle}.$$

7.2.2 Why orthogonality matters

Lemma 7.2.5. Orthogonality guarantees linear independence

Main Idea / Strategy

Formal Proof

Suppose that some finite combination of orthogonal functions is zero,

$$\sum_{n=0}^{N} c_n \psi_n(x) = 0.$$

Taking the weighted inner product with ψ_m gives

$$\sum_{n=0}^{N} c_n \langle \psi_n, r(x)\psi_m \rangle = c_m \langle \psi_m, r(x)\psi_m \rangle = 0.$$

Since ψ_m is non-zero,

$$\langle \psi_m, r(x)\psi_m \rangle > 0,$$

and hence

$$c_m = 0.$$

As this is true for every m, all of the coefficients must vanish. Thus an orthogonal collection of non-zero functions is linearly independent. □

Remark 7.2.6 (Why does linear independence matter?)**.** It means that the different basis functions describe genuinely separate components of the function. One component cannot be recreated by combining the others. This is the fundamental idea behind the use of function expansions in this course: we describe a complicated function by the behaviour of many simple, independent functions.

Proposition 7.2.7 (Linear independence lets the dynamics separate)**.** Later, when we study partial differential equations, this becomes extremely powerful. We will write a solution in the form

$$u(x,t) = \sum_{n=0}^{\infty} c_n(t) y_n(x).$$

For an appropriate linear PDE, substitution will lead to an expression of the form

$$\sum_{n=0}^{\infty} F_n(t) y_n(x) = 0.$$

Because the functions $y_n(x)$ are orthogonal (and hence independent), we can isolate each component by taking the weighted inner product with y_m :

$$\left\langle \sum_{n=0}^{\infty} F_n(t) y_n, r(x) y_m \right\rangle = F_m(t) \langle y_m, r(x) y_m \rangle = 0.$$

Since $\langle y_m, r(x) y_m \rangle > 0$, it follows that

$$F_m(t) = 0,$$

for every m, giving us a separate ordinary differential equation for each coefficient $c_m(t)$.So, schematically,

one PDE for a complicated function $\rightarrow$ ODEs for simple independent components.

This is one of the main reasons function-basis expansions are so useful in applied mathematics.

Remark 7.2.8. We can decompose complex functions into a summation of simpler functions that allow for direct analysis of behaviour.

7.2.3 An infinite dimensional basis

Remark 7.2.9. For an infinite expansion, we require more than linear independence to ensure a unique representation. While this is fine in a Finite expansion, it is not for infinite. Instead, we require the collection of functions to be complete.

Definition 7.2.10 (Complete)**.** A collection is complete if all functions in the collection can be represented with the modes.

Theorem 7.2.11 (Sturm-Liouville Completeness Theorem)**.** Consider the weighted Sturm-Liouville eigenvalue problem

$$Ly = \lambda r(x) y,$$

where

$$Ly = \frac{\mathrm{d}}{\mathrm{d}x}\left(p(x)\frac{\mathrm{d}y}{\mathrm{d}x}\right) + q(x)y, \quad a \leq x \leq b,$$

with separated homogeneous boundary conditions

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0,$$
$$\alpha_3 y(b) + \alpha_4 y'(b) = 0.$$

For a regular Sturm-Liouville problem, the eigenfunctions

$$y_0(x), y_1(x), y_2(x), \ldots$$

form a complete orthogonal set with weight $r(x)$.

Thus, for any

$$f \in L^2_r([a,b]), \quad \int_a^b |f(x)|^2 r(x)\mathrm{d}x < \infty,$$

we can write, with convergence in the weighted L^2 sense,

$$f(x) = \sum_{n=0}^{\infty} c_n y_n(x),$$

where the coefficients are uniquely determined by

$$c_n = \frac{\int_a^b f(x) y_n(x) r(x)\mathrm{d}x}{\int_a^b y_n^2(x) r(x)\mathrm{d}x}.$$

Remark 7.2.12. There are therefore three ingredients working together:

1. Orthogonality lets us isolate each coefficient.
2. Orthogonality implies linear independence, so the modes describe genuinely separate components.
3. Completeness tells us that there are enough of these independent components to represent every function in the relevant function space.

7.2.4 Extension to higher dimensions

Theorem 7.2.13 (Results extend to higher dimensions)**.** Let

$$\mathbf{x} = (x_1, x_2, \ldots, x_n) \in D \subset \mathbb{R}^n,$$

and consider functions $u(\mathbf{x})$ and $v(\mathbf{x})$ defined on D. The inner product is

$$\langle u, v\rangle = \int_D u(\mathbf{x})v(\mathbf{x})\mathrm{d}^n\mathbf{x},$$

The corresponding weighted norm is

$$\|u\|^2 = \langle u, r(\mathbf{x})u\rangle = \int_D r(\mathbf{x})|u(\mathbf{x})|^2 \ \mathrm{d}^n\mathbf{x}.$$

where $r(\mathbf{x}) > 0$ is an appropriate weight function. Exactly as in one dimension, two functions are orthogonal with weight $r(\mathbf{x})$ if

$$\langle u, rv\rangle = 0.$$

Suppose we have a complete set of orthogonal basis functions

$$\{\Phi_{\mathbf{k}}(\mathbf{x})\},$$

where

$$\mathbf{k} = (k_1, k_2, \ldots, k_n)$$

is a collection of integers labelling a particular mode. A function can then be expanded as

$$f(\mathbf{x}) = \sum_{\mathbf{k}} a_{\mathbf{k}}\Phi_{\mathbf{k}}(\mathbf{x}).$$

The important point is that the coefficient grab is exactly the same as in one dimension:

$$a_{\mathbf{k}} = \frac{\langle f, r(\mathbf{x})\Phi_{\mathbf{k}}\rangle}{\langle \Phi_{\mathbf{k}}, r(\mathbf{x})\Phi_{\mathbf{k}}\rangle}.$$

Separable Basis

Remark 7.2.14. We can write certain domains basis functions as the product of one dimensional basis functions.For example, suppose

$$D = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n],$$

and the basis functions have the form

$$\Phi_{\mathbf{k}}(\mathbf{x}) = \psi_{k_1}^{(1)}(x_1)\,\psi_{k_2}^{(2)}(x_2)\cdots\psi_{k_n}^{(n)}(x_n)\,.$$

If the weight also separates,

$$r(\mathbf{x}) = r_1(x_1)\,r_2(x_2)\cdots r_n(x_n)\,,$$

then the inner product of two basis functions separates into a product of one-dimensional inner products:

$$\langle \Phi_{\mathbf{k}}, r(\mathbf{x})\Phi_{\mathbf{m}}\rangle = \prod_{j=1}^{n}\left\langle \psi_{k_j}^{(j)}, r_j(x_j)\,\psi_{m_j}^{(j)}\right\rangle.$$

Hence orthogonality in each coordinate direction produces orthogonality of the multidimensional modes.